

RESEARCH ARTICLE

10.1002/2015JG003247

Key Points:

- Long-term year-round measurements of CH₄ and CO₂ fluxes from rice agriculture
- Interannual variability in CH₄ and CO₂ fluxes was largely driven by temperature
- Temperature also influenced the ratio of CH₄ to CO₂ emissions to across diel to interannual scales

Supporting Information:

- Supporting Information S1

Correspondence to:

S. H. Knox,
saraknox.knox@gmail.com

Citation:

Knox, S. H., J. H. Matthes, C. Sturtevant, P. Y. Oikawa, J. Verfaillie, and D. Baldocchi (2016), Biophysical controls on interannual variability in ecosystem-scale CO₂ and CH₄ exchange in a California rice paddy, *J. Geophys. Res. Biogeosci.*, 121, 978–1001, doi:10.1002/2015JG003247.

Received 14 OCT 2015

Accepted 28 FEB 2016

Accepted article online 30 MAR 2016

Published online 31 MAR 2016

Corrected 7 MAY 2016

This article was corrected on 7 MAY 2016. See the end of the full text for details.

Biophysical controls on interannual variability in ecosystem-scale CO₂ and CH₄ exchange in a California rice paddy

Sara Helen Knox¹, Jaclyn Hatala Matthes², Cove Sturtevant³, Patricia Y. Oikawa¹, Joseph Verfaillie¹, and Dennis Baldocchi¹
¹Ecosystem Science Division, Department of Environmental Science, Policy and Management, University of California, Berkeley, California, USA, ²Department of Geography, Dartmouth College, Hanover, New Hampshire, USA, ³Fundamental Instrument Unit, National Ecological Observatory Network, Boulder, Colorado, USA

Abstract We present 6.5 years of eddy covariance measurements of fluxes of methane (F_{CH_4}) and carbon dioxide (F_{CO_2}) from a flooded rice paddy in Northern California, USA. A pronounced warming trend throughout the study associated with drought and record high temperatures strongly influenced carbon (C) budgets and provided insights into biophysical controls of F_{CO_2} and F_{CH_4} . Wavelet analysis indicated that photosynthesis (gross ecosystem production, GEP) induced the diel pattern in F_{CH_4} , but soil temperature (T_s) modulated its amplitude. Forward stepwise linear models and neural networking modeling were used to assess the variables regulating seasonal F_{CH_4} . As expected due to their competence in modeling nonlinear relationships, neural network models explained considerably more of the variance in daily average F_{CH_4} than linear models. During the growing season, GEP and water levels typically explained most of the variance in daily average F_{CH_4} . However, T_s explained much of the interannual variability in annual and growing season CH₄ sums. Higher T_s also increased the annual and growing season ratio of F_{CH_4} to GEP. The observation that the F_{CH_4} to GEP ratio scales predictably with T_s may help improve global estimates of F_{CH_4} from rice agriculture. Additionally, T_s strongly influenced ecosystem respiration, resulting in large interannual variability in the net C budget at the paddy, emphasizing the need for long-term measurements particularly under changing climatic conditions.

1. Introduction

Wetland rice paddies are a major source of atmospheric methane (CH₄) [Reeburgh, 2003; Whalen, 2005], the second most important greenhouse gas following carbon dioxide (CO₂) [Myhre et al., 2013]. Numerous studies over the past decades have measured CH₄ emissions from rice fields throughout the world [cf. Aulakh et al., 2001; Yan et al., 2005]. Previous studies have identified a number of factors and processes controlling CH₄ emissions, including soil properties [Huang et al., 1997a; Neue and Sass, 1994; Sass et al., 1994; Yagi and Minami, 1990], agricultural practices such as floodwater management [Alberto et al., 2014; Corton et al., 2000; Linquist et al., 2015; Wassmann et al., 2000a; Yagi et al., 1996] and fertilizer application [Bronson et al., 1997; Cicerone et al., 1992; Corton et al., 2000; Lindau et al., 1991; Naser et al., 2007; Sass et al., 1991a; Schütz et al., 1989a; Wassmann et al., 2000a; Yagi and Minami, 1990], cultivar selection [Ding et al., 1999; Huang et al., 1997a; Wassmann and Aulakh, 2000], grain to biomass ratios [Denier van der Gon et al., 2002], and climate [Neue and Sass, 1994; Sass et al., 1991b; Schütz et al., 1990; Wassmann et al., 2000b; Yan et al., 2005]. In addition, CH₄ is strongly regulated by the ecological function of rice plants since they are the primary transport mechanisms of CH₄ from the soil to the atmosphere with diffusion being the dominant mechanism for gas exchange in rice [Chanton et al., 1997; Den van der Gon and Van Breemen, 1993; Wassmann and Aulakh, 2000]. Rice plants also promote CH₄ oxidation via transport of atmospheric oxygen through aerenchyma, and they are an important source of carbon (C) substrates for methanogenic metabolism [Cicerone et al., 1992; Hatala et al., 2012a; Huang et al., 1997b; Wassmann and Aulakh, 2000]. Even in wetland ecosystems, where soil organic carbon can exceed 90%, there is strong evidence to suggest that CH₄ production is fueled by recent plant photosynthates in the form of exudates in the rhizosphere with limited contributions from recalcitrant soil organic matter [Bridgman et al., 2013; Ström et al., 2003]. However, considerable uncertainty remains regarding the quantitative relationships between CH₄ emissions from rice paddies and the biophysical processes driving these fluxes [Huang et al., 1997a], which contributes to the large uncertainty (25 to 300 Tg CH₄ yr^{−1}, with a median value of 53 Tg CH₄ yr^{−1}) in global CH₄ budgets from rice agriculture [Bridgman et al., 2013].

Methane emissions from rice fields have predominantly been measured using the closed chamber technique [cf. Aulakh *et al.*, 2001]. While chambers are advantageous for assessing spatial variability and treatment effects on fluxes, are easy to manipulate, and relatively inexpensive, placing a chamber over the plant-soil environment can introduce a number of potential biases due to direct interaction with the near-surface environment [Baldocchi, 2003; Chanton *et al.*, 1997; Meijide *et al.*, 2011]. Furthermore, measurements are performed over a limited area and are often discrete in time, which provides challenges for estimating robust annual budgets since CH₄ emissions from rice show large spatiotemporal variability [Holzapfel-Pschorn and Seiler, 1986; Khalil and Butenhoff, 2008; Schütz *et al.*, 1989a]. In recent years, the eddy covariance technique has emerged as an alternative means of measuring trace gas exchange since it provides quasi-continuous measurements at the ecosystem scale without interfering with the processes of gas exchange between the surface and the atmosphere [Baldocchi, 2003]. While the eddy covariance method has largely been used to measure CO₂, water, and energy fluxes, the development of open-path sensors suitable for remote and fast CH₄ measurements [Detto *et al.*, 2011; McDermitt *et al.*, 2011] has allowed a growing number of studies to measure CH₄ fluxes using this method [e.g., Baldocchi *et al.*, 2012; Chu *et al.*, 2014; Hatala *et al.*, 2012b; Hendriks *et al.*, 2010, 2007; Herbst *et al.*, 2013; Knox *et al.*, 2014; Koebsch *et al.*, 2015; Kroon *et al.*, 2010; Matthes *et al.*, 2014; Petrescu *et al.*, 2015; Rinne *et al.*, 2007; Sturtevant and Oechel, 2013; Sturtevant *et al.*, 2012]. However, to date few studies have used this method to measure CH₄ emissions from flooded rice paddies on seasonal or annual timescales [Alberto *et al.*, 2014; Bhattacharyya *et al.*, 2014; Hatala *et al.*, 2012a, 2012b; Knox *et al.*, 2014; Meijide *et al.*, 2011].

Concurrent near-continuous measurements of CH₄, CO₂, and water vapor, along with ancillary physical variables (e.g., temperature, water depth, and radiation), can improve our understanding of the major controlling factors of CH₄ emission across a spectrum of timescales (hours to years) [Baldocchi, 2014; Chu *et al.*, 2014; Koebsch *et al.*, 2015]. Considerable insight can be gained by exploring which of these variables are strongly correlated with CH₄ exchange and under what conditions, even if these variables, such as ecosystem respiration of CO₂ or ecosystem photosynthesis and evapotranspiration, may not represent direct drivers of CH₄ production, oxidation, or transport processes [Morin *et al.*, 2014; Sturtevant *et al.*, 2015]. For example, previous studies observed tight couplings between rice photosynthesis and CH₄ exchange across periods of hours [Hatala *et al.*, 2012a] to seasons [Chanton *et al.*, 1997; Huang *et al.*, 1997b; Sass *et al.*, 1990], highlighting the connection between recent photosynthates and CH₄ production and/or the influence of stomatal conductance on CH₄ transport. Additionally, exploring the relationship between evapotranspiration and CH₄ flux can provide insight into the influence of plant activity and vegetation dynamics on CH₄ transport [Morin *et al.*, 2014; Sturtevant *et al.*, 2015], including the role of transpiration in driving diel variations in CH₄ emission [Chanton *et al.*, 1997]. Ecosystem respiration has also been found to be an important predictor of CH₄ flux, improving the prediction of CH₄ emissions beyond the contribution of other direct environmental drivers alone, highlighting the complex and nonlinear relationships of these environmental drivers on metabolic processes across the microbial community [Morin *et al.*, 2014]. Simultaneous measurements of CO₂ and CH₄ fluxes from rice fields over annual time periods also allow the quantification of CO₂ to CH₄ ratios that can be used to assess the potential of estimating the flux of one gas from the other [McMillan *et al.*, 2007; Stallard, 1998; Whiting and Chanton, 1993] and the extent to which net CO₂ uptake can offset the radiative effect of CH₄ emissions, which currently remain uncertain since only a limited number of studies have simultaneously quantified annual budgets of both CH₄ and CO₂ from rice agriculture [Bhattacharyya *et al.*, 2014; Hatala *et al.*, 2012b; McMillan *et al.*, 2007].

In this paper we present results from a 6.5 year study of simultaneous measurements of CH₄, CO₂, and water fluxes from a flooded rice field in Northern California using the eddy covariance method. To our knowledge, this is the longest and most comprehensive data set of eddy covariance measurements of wetland rice greenhouse gas fluxes in the world. Our objectives were to investigate factors affecting CH₄ fluxes across diel to interannual timescales and quantify interannual variability in CO₂ and CH₄ budgets. Specifically, we addressed the following questions: (1) How do CH₄ fluxes at the rice paddy vary across diel, seasonal, and interannual timescales? (2) What is the net C balance and CO₂ to CH₄ ratio of the site and how does this vary from year to year? And (3) what biophysical factors are important for predicting CH₄ fluxes and how do these factors vary across timescales? The severe drought and associated record high temperatures experienced in California during the latter half of the study produced strong variability in environmental drivers, which provided a broad range of conditions to investigate the biophysical regulators of CO₂ and CH₄ exchange in wetland rice systems.

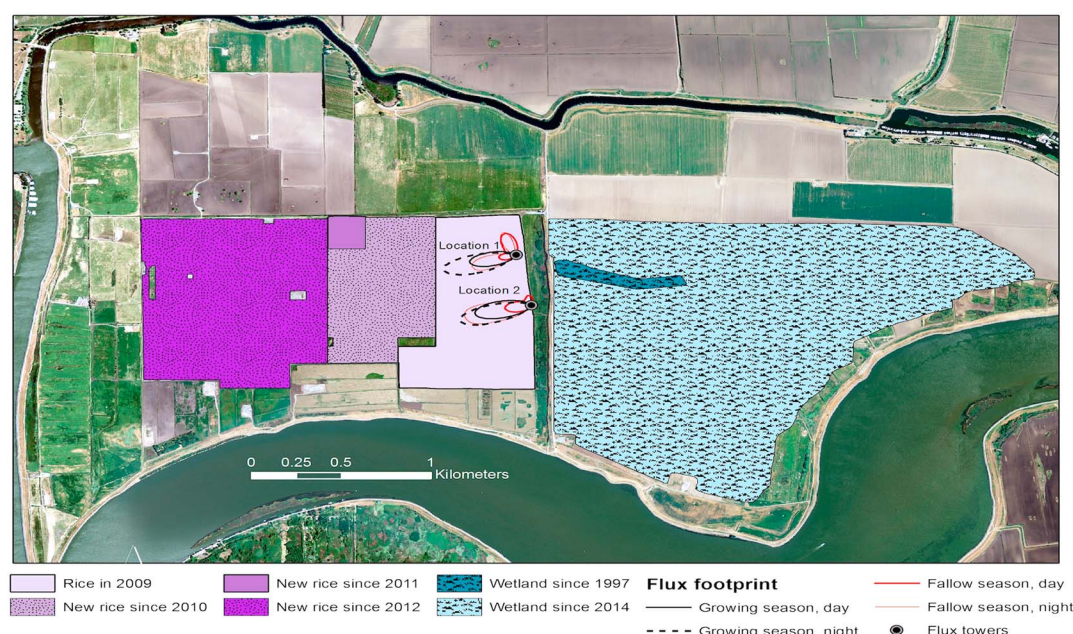


Figure 1. Flux tower location, climatological footprint, and extent of flooded rice and wetlands on Twitchell Island, CA, USA, throughout the study period. The tower was primarily located at Location 1 throughout the study (38.10875°N, 121.6530°W), except from 22 July 2009 to 17 November 2010 when it was moved 300 m to the south (Location 2; 38.105530°N, −121.652097°W). For the climatological footprint, data are separated by season (growing season versus fallow season), and day and night status and contours represent the 85% flux footprints. The extent of rice on the island varied throughout the study, and by 2014, the area of flooded rice and wetlands on the island approached 6 km².

2. Materials and Methods

2.1. Study Site and Crop Management

Our study site is located on Twitchell Island in the Sacramento-San Joaquin Delta, California, USA (referred to hereafter as the Delta) (Figure 1), which is located approximately 100 km inland from the Pacific Ocean. The region experiences a Mediterranean-type climate characterized by hot, dry summers and cool, wet winters. The 18 year average temperature (1998–2015) recorded at a nearby weather station located ~1 km northwest of our site on Twitchell Island is 15.1°C and mean annual precipitation is 326 mm. The Delta exists as a network of islands that maintain a water table below sea level through an extensive levee network and pumping [Deverel and Leighton, 2010; Mount and Twiss, 2005] following the drainage of the Delta's tidal marshes over a century ago. Agricultural cultivation since drainage has caused high rates of peat soil oxidation, where today the land surface in the Delta lies up to 8 m below sea level [Deverel and Rojstaczer, 1996; Drexler et al., 2009].

Prior to 1990 rice was not farmed in the Delta due to the lack of appropriate cultivars for the cool nighttime growing season temperatures. However, with the development of new varieties and the potential for rice agriculture to slow subsidence in the Delta by lowering rates of ecosystem respiration under flooded conditions [Hatala et al., 2012b; Knox et al., 2014], today, there are about 20 km² of rice farmed in the region. The Twitchell Island rice paddy is a pilot project managed by the California Department of Water Resources, and the land was converted from traditional drained corn and alfalfa to rice (*Oryza sativa*) in 2009. Figure 1 shows the progressive increase in the size of flooded rice agriculture and wetlands on Twitchell Island throughout the study. During the first year, the east-west fetch was less than 500 m; however, by 2014 rice covered 2.44 km² of land and the total area of flooded vegetation was 5.7 km² (Figure 1). As described by D. D. Baldocchi et al. (The impact of expanding flooded land area on the annual evaporation of rice, submitted to *Agricultural and Forest Meteorology*, 2016), during the first few years of the study, the small and relatively isolated extent of flooded rice promoted an “oasis effect” that enabled warm dry air to be entrained across the top of the planetary boundary layer. However, by the sixth year of the study, the area of the flooded rice and wetlands approached a horizontal scale that appeared to inhibit the oasis effect. As suggested from cospectral analysis and analysis with a coupled surface energy balance planetary boundary model, this likely resulted in water and heat advection in the earlier years of the study (D. D. Baldocchi et al., submitted manuscript, 2016).

Table 1. Cultivars Planted, Planting, Harvest, Fertilizer and Herbicide Application Dates, Peak Plant Area Index (PAI), Yield at 14% Moisture Content, and Grain Carbon

	2009–2010	2010–2011	2011–2012	2012–2013	2013–2014	2014–2015	2015–2016
Variety	M104	M206	M104	M104	M206	M206	M206
Plant date	15 Apr	16 Apr	22 Apr	17 May	2 Apr	28 Apr	16 Apr
Harvest date	20–25 Oct	28–29 Oct	12–14 Oct	13–15 Nov	20 Sep	2 Oct	23 Sep
Fertilizer application	15 Apr 100 lb ac ⁻¹ of 11-52-0	16 Apr 100 lb ac ⁻¹ of 11-52-0	22 Apr 100 lb ac ⁻¹ of 11-52-0	17 May 100 lb ac ⁻¹ of 11-52-0	2 May 100 lb ac ⁻¹ of 15-15-15	5 Jun 150 lb ac ⁻¹ of 15-15-15	31 May 100 lb ac ⁻¹ of 15-15-15
Herbicide application	23–24 May 150 lb ac ⁻¹ of 30-0-20 22 May Bispyribac sodium and Pendimethalin	5–9 Jun 150 lb ac ⁻¹ of 30-0-20 8 Feb Alecto, Mist Control, Sharl EW 3 Jun Regiment, Prowl, Syl-Tac Uan-32, Sandeia	14 Jun 200 lb ac ⁻¹ of 30-0-20 10 Jun Regiment, Prowl, Syl-Tac Uan-32, Sandeia 5 Jul Sprayed for water grass from helicopter	16 Jun 200 lb ac ⁻¹ of 30-0-20 31 May Regiment, Prowl, Syl-Tac Uan-32, Sandeia 6 Jun Clincher CA, Mor-Act Adjuvant, Tri-Fol 19 Jul SuperWham	1 May Regiment, Prowl, Syl-Tac Uan-32, Sandeia	4 Jun Regiment, Prowl, Syl-Tac Uan-32, Sandeia	1 Jun Regiment, Prowl, Syl-Tac Uan-32, Sandeia 25 Jun SuperWham
Peak PAI	6.7	7	4.6	4.1	4.6	5	3.5
Yield (t ha ⁻¹)	4.7	4.5	6.0	5.6	7.3	5.7	7.4
Grain carbon ^a (g C m ⁻²)	172	163	220	205	266	208	269

^aCalculated from yields as in Hollinger *et al.* [2005] assuming that the fraction of carbon in the grain is 0.43 based on measurements.

However, the cospectral densities for $\overline{w'c'}$ and $\overline{w'm'}$ overlapped one another during 2009 (when there was an oasis effect) and 2014 (when there was no evidence for an oasis effect), suggesting that there was likely no CO₂ or CH₄ advection and therefore minimal influence on CO₂ and CH₄ fluxes and budgets.

The rice paddy lies ~5 m below sea level and is located on degraded peat soil with the topmost layer consisting of a silt loam which overlays a deep peat layer. The soil is characterized by low bulk density (0.65 and 0.57 g cm⁻³ at a depth of 0–30 cm and 30–45 cm below the soil surface, respectively) and high C content (15% and 31% at a depth of 0–30 cm and 30–45 cm below the soil surface, respectively) [Hatala *et al.*, 2012b]. Crop management practices are summarized in Table 1 and varied somewhat from 2009 to 2015 as changes were made in attempts to improve yields. The field was typically disced once or twice and ring rolled for planting in late-March to mid-April, with the exception of the 2012 growing season when planting was delayed until mid-May due to late rainy season precipitation. Rice varieties M104 or M206, both semidwarf, cold weather cultivars were planted at a density of 14–17 g m⁻². Between 2009 and 2012 the field was fertilized both at planting and just before flooding; however, from 2013 onward fertilizer was only applied once shortly before flooding (Table 1). The site was treated with herbicide one to three times throughout the growing season depending on weed infestations (Table 1). The field was flooded throughout the growing season and harvested between late September and late October. With the exception of 2014, the rice straw was chopped and left on the field after harvesting of the grain. The paddy was reflooded throughout the winter to provide habitat for migrating birds and reduce rates of peat soil oxidation. Few plants grew within the field during the flooded winter period, and therefore, wintertime photosynthesis represented typically <10% of annual gross photosynthetic uptake.

2.2. Eddy Covariance Measurements

We used the eddy covariance method to measure 30 min average fluxes of CH₄ (F_{CH_4} ; nmol m⁻² s⁻¹), CO₂ (F_{CO_2} ; μmol m⁻² s⁻¹), latent heat (LE; W m⁻²), and sensible heat (H ; W m⁻²) [Baldocchi *et al.*, 1988]. The eddy covariance instrumentation was mounted on a tower at an average height of 3.15 m. The tower was primarily located near the northeastern portion of the field (Location 1), except from 22 July 2009 to 17 November 2010 when it was moved 300 m to the south (Location 2). As the field is homogeneous, the location of the

tower is likely to have minimal influence on ecosystem-scale F_{CO_2} and F_{CH_4} . High-frequency three-dimensional wind speed (u, v, w ; m s^{-1}) and temperature (T_{sonic}) were measured with a sonic anemometer (Gill WindMaster or WindMaster Pro; Gill Instruments Ltd, Lymington, Hampshire, England). Fluctuations in CO_2 and H_2O molar density were measured with an open-path infrared gas analyzer (LI-7500; LI-COR Biogeosciences, Lincoln, NE, USA). Prior to 17 January 2012, CH_4 mixing ratio (χ_{CH_4}) was measured with a closed-path tunable diode laser CH_4 analyzer (Los Gatos Research (LGR), Mountain View, CA, USA). Two types of LGR analyzers were used: a fast greenhouse gas analyzer (FGGA) that simultaneously measures concentrations of CO_2 , CH_4 , and H_2O (4 June 2009 to 17 August 2010) and a fast methane analyzer (FMA) (17 August 2010 to 17 January 2012). Following 17 January 2012 an open-path CH_4 analyzer, based on wavelength modulation spectroscopy, was used to measure fluctuations in CH_4 molar density (LI-7700; LI-COR Biogeosciences, Lincoln, NE, USA). The specifications and details of the external pump required to operate the closed-path system are given in *Detto et al.* [2011] and *Hatala et al.* [2012b]. Extensive field testing was done to evaluate the performance of the closed-path CH_4 sensors at this site [*Detto et al.*, 2010], and good agreement was observed in the magnitude of 30 min fluxes between the open-path and closed-path flux systems [*Detto et al.*, 2011]. Raw turbulence data were initially recorded with a Campbell CR1000 data logger (Campbell Scientific, Logan, UT, USA) and following 7 May 2012 using a digital data logger system (LI-7550A; LI-COR Biogeosciences, Lincoln NE, USA). Prior to 27 July 2013 anemometer readings and trace gas densities were recorded at 10 Hz intervals, but thereafter, the data were collected at 20 Hz since it resulted in slightly less high-frequency attenuation.

Fluxes were calculated using the 30 min covariance of vertical wind speed (w) and scalars of interest after applying a series of standard corrections using in-house software [*Detto et al.*, 2010; *Hatala et al.*, 2012b] as outlined in *Knox et al.* [2014]. Briefly, these included despiking high-frequency data, applying coordinate rotations, correcting for effects of air density fluctuations, filtering fluxes with low friction velocity (0.12 to 0.15 m s^{-1} , varying somewhat seasonally) and spikes in 30 min mean densities, variances and covariances, and rejecting fluxes from winds between 0° and 190° for Location 1 and outside of 200° – 350° for Location 2, which was estimated based on the relationship between wind direction and fluxes, and the flux footprint calculated using a 2-D footprint model (Figure 1). Details of the 2-D footprint model can be found in the supporting Supporting Information S1. Unlike previous versions of the in-house software that used a default time lag between w and scalars defined by the user, a variable lag was calculated based on a maximum cross-correlation method. The software was also updated to filter fluxes with large vertical rotation angles ($> \pm 7^\circ$) and flagged fluxes using the stationarity test of *Foken and Wichura* [1996]. The percentage of 30 min fluxes excluded annually from this analysis, including periods of power loss and sensor malfunction, ranged between 43 and 54% for F_{CO_2} and 44 and 61% for F_{CH_4} with the exception of the first year of measurement when 82% of the data were excluded due to the late deployment of the CH_4 analyzer (June) and generator outages and sensor malfunction. However, much less data were excluded during the growing season (13–32% for F_{CO_2} and 13–38% for F_{CH_4} again with the exception of 2009 when 77% of the data were excluded) due to the prevalence of clear sunny days and robust and persistent winds that flow through the Carquinez Strait between the cool coast and the warm interior valley (Figure 1).

No high-frequency corrections were applied to fluxes computed using open-path sensors since cospectral analysis indicated that flux losses were less than 5% [*Knox et al.*, 2014], which is well within the accuracy of individual eddy covariance flux measurements and of the correction algorithms [*Aubinet et al.*, 1999; *Massman and Lee*, 2002; *Moore*, 1986]. However, corrections for the closed-path CH_4 measurements were applied to account for the high-frequency flux losses due to sensor separation and attenuation of χ_{CH_4} fluctuations down the sampling tube. When the FGGA was present, corrections were calculated as in *Detto et al.* [2011] by taking advantage of the additional CO_2 channel. When the FMA sensor was present and we could not take advantage of the additional CO_2 channel, we used a procedure similar to the one outlined in *Aubinet et al.* [1999] and *Humphreys* [2004] as described in *Knox et al.* [2014]. F_{CH_4} corrected for spectral loss agreed to within about 3% with CH_4 emissions measured using an open-path sensor for periods when both systems were present for comparison (17 August 2010 to 21 September 2010 and 17 January 2012 to 25 April 2012).

Energy balance closure, defined as the slope between daily net radiation and the residual of the energy balance [*Wilson et al.*, 2002], was 0.84 for the entire study period and ranged between 0.81 and 0.89 when calculated for individual years. In addition to the ground heat flux, storage terms included radiation energy absorbed in photosynthesis calculated as in *Leuning et al.* [2012] and heat storage in the water column

approximated as in Drexler *et al.* [2004]. Energy balance closure reported in this study falls within the range typically observed at sites within the FLUXNET network [Stoy *et al.*, 2013; Wilson *et al.*, 2002], and on the upper end of values reported for wetland sites [Malone *et al.*, 2014; Stoy *et al.*, 2013], where lower closure is typically observed due to the difficulties in estimating storage terms [Malone *et al.*, 2014].

2.3. Gap-Filling, Flux Partitioning, and Budget Estimations

Half-hourly fluxes were gap-filled using an artificial neural network (ANN) method [Baldocchi and Sturtevant, 2015; Dengel *et al.*, 2013; Knox *et al.*, 2014; Moffat *et al.*, 2007; Morin *et al.*, 2014; Papale *et al.*, 2006]. To facilitate representativeness, explanatory data were divided into a maximum of 20 natural data clusters using the *k*-means method. To avoid biasing toward conditions with better flux data coverage, data used to train, test, and validate the ANN were proportionately sampled from these clusters. Several architectures of increasing complexity were tested to facilitate generalizability of the neural network. The architecture of each neural network was initialized 10 times with random starting weights, and the initialization resulting in the lowest mean sampling error was used. The simplest architecture, whereby additional increases in complexity resulted in <5% reduction in mean square error, was selected and the prediction saved. This procedure was replicated with 20 resamplings of the data, and the median prediction was used to fill missing half hours.

While ecosystem photosynthesis (gross ecosystem production; GEP) and respiration (ER) are not directly measured via the eddy covariance method, net CO₂ fluxes (F_{CO_2}) can be partitioned into GEP and ER by separately considering daytime and nighttime observations because photosynthesis only occurs during daylight hours. Since winds in the Delta are strong even at night [Baldocchi and Sturtevant, 2015; Hatala *et al.*, 2012b; Knox *et al.*, 2014], uncertainties in nighttime eddy covariance measurements due to atmospheric stratification and stability are minimized [Massman and Lee, 2002]. F_{CO_2} was gap-filled separately for daytime and nighttime observations. Predictions from the ANN resulting from the nighttime gap-filling were used to model ER for all data (daytime and nighttime values), and GEP was calculated by subtracting gap-filled F_{CO_2} from modeled respiration ($F_{\text{CO}_2} = \text{ER} - \text{GEP}$) [Baldocchi and Sturtevant, 2015]. In this study, negative F_{CO_2} values indicated net CO₂ uptake while positive values indicate net CO₂ release, while both GEP and ER are presented with positive signs.

For half-hourly fluxes, the entire observation period was used to train, test, and validate the ANNs. Explanatory variables used to gap-fill daytime F_{CO_2} included air temperature (T_a), photosynthetically active radiation (PAR), vapor pressure deficit (VPD), water table depth (WTD), friction velocity (u^*), decimal day of year since the start of the study, and sine and cosine functions to represent seasonal changes. Explanatory variables used to gap-fill nighttime F_{CO_2} and partition ER included the same variables as daytime F_{CO_2} with the exception of PAR and VPD. Unlike F_{CO_2} , LE and F_{CH_4} were not gap-filled separately for daytime and nighttime data. Explanatory variables used to gap-fill LE were the same as daytime F_{CO_2} , and variables used to gap-fill F_{CH_4} included soil temperature (T_s) at 2 cm depth, WTD, u^* , GEP, ER, LE, decimal day of year, and sine and cosine functions, which represent the main environmental drivers of CH₄ emissions or variables that strongly covary with F_{CH_4} [Chu *et al.*, 2014; Morin *et al.*, 2014; Sturtevant *et al.*, 2015; Treat *et al.*, 2007]. The optimal neural network for both daytime and nighttime F_{CO_2} gap-filling included two hidden layers with 15 and 8 hidden nodes, respectively, while most often the optimal neural network for F_{CH_4} gap-filling included two hidden layers with 10 and 5 hidden layers, respectively.

Gap-filling and partitioning routines performed very well; daytime (nighttime) neural network predictions for F_{CO_2} gave a model r^2 of 0.95 (0.76) with a RMSE of 2.4 (1.5) $\mu\text{mol m}^{-2} \text{s}^{-1}$, for F_{CH_4} model r^2 was 0.79 with a RMSE of 30 $\text{nmol m}^{-2} \text{s}^{-1}$, and for LE model r^2 was 0.93 with a RMSE of 0.82 $\text{mmol m}^{-2} \text{s}^{-1}$. We also tested the Reichstein *et al.* [2005] algorithm to estimate ER and found negligible differences in the results presented below while annual budgets of GEP and ER differed on average <10% between the two flux partitioning methods.

Annual and seasonal sums of F_{CO_2} , GEP, ER, and F_{CH_4} were calculated by integrating gap-filled and partitioned fluxes over time. For annual budgets, half-hourly fluxes were integrated from 1 April to 1 April of the following year. For figures, labels refer to the year that includes the growing season (e.g., 2009 is used as the label for 2009–2010). Growing season budgets were calculated by integrating fluxes from planting to harvest of a given year, while fallow season budgets were calculated from harvest of 1 year to planting of the following year. To estimate the amount of uncertainty in the ANN gap-filling procedure for the annual and seasonal budgets, we calculated the integrated budget using the full range of predictions used to fill each gap (i.e., based on the 20 extractions from the ANN) and calculated the 95% confidence interval from the distribution of the 20 annual or seasonal budgets [Knox *et al.*, 2014].

2.4. Meteorological and Vegetation Measurements

Meteorological conditions including T_a , relative humidity, WTD, incoming solar radiation, net radiation (R_{net}), PAR, ground heat flux, T_s at depths of -0.02 , -0.04 , -0.08 , -0.16 , and -0.32 m below the ground surface, and water temperature (T_w) were characterized with a standard set of sensors as described in Knox *et al.* [2014]. Gaps in T_a were filled using data from a nearby weather station on Twitchell Island, while gaps in other meteorological variables were filled with a 2-D interpolator that conserves the mean diurnal variation. There were few gaps in meteorological measurements (e.g., $<1\%$ for PAR for the entire observation period); however, ground sensors were removed from the soil for approximately 2 weeks to a month for each planting and harvesting event.

Plant area index (PAI) was measured approximately weekly or biweekly during the growing season. Measurements were made using a LAI-2000 Plant Canopy Analyzer (LI-COR Biogeosciences, Lincoln, NE, USA). Measurements were recorded every 10 m along a 100 m transect extending west, the predominant wind direction at the site. The fraction of photosynthetically active radiation (FPAR) absorbed by the canopy was calculated from PAI using the Beer-Lambert law [Ruimy *et al.*, 1999] assuming a light extinction coefficient of 0.6 [Monteith, 1969]. FPAR was then multiplied by PAR to estimate absorbed PAR. Canopy phenology and crop management were monitored with a digital camera [Richardson *et al.*, 2007].

2.5. Data Analysis

We used a combination of linear regression models, neural network modeling, and time series analysis to investigate the biophysical controls on F_{CH_4} from the rice paddy. Different data analysis techniques were used for each timescale of interest, taking advantage of the analysis technique most suitable to each timescale.

2.5.1. Interannual and Seasonal Timescales

Due to the limited number of data points ($n=6$ or 7 years or growing seasons, respectively), pairwise linear regression models were used to explore the factors predicting CH_4 fluxes at the interannual scale. At the seasonal scale, we took advantage of the significantly larger number of daily average flux and environmental variables to explore neural network models [Dengel *et al.*, 2013; Knox *et al.*, 2014; Morin *et al.*, 2014], in addition to linear models, for daily average F_{CH_4} (referenced simply as “daily F_{CH_4} ” hereafter). We used a statistical model selection process similar to Morin *et al.* [2014] to assess the effectiveness of T_a , T_s , WTD, u^* , GEP, ER, and LE in predicting daily F_{CH_4} since these are the primary variables regulating or covarying with F_{CH_4} [Chu *et al.*, 2014; Koebisch *et al.*, 2015; Morin *et al.*, 2014; Sturtevant *et al.*, 2015; Treat *et al.*, 2007]. Both environmental variables and fluxes of carbon and water were included as potential drivers of F_{CH_4} because they can represent strongly shared mechanisms between F_{CH_4} and other environmental or flux variables even if they are not directly linked by causality, and thus, their inclusion can provide empirical predictive power for F_{CH_4} [Morin *et al.*, 2014].

We divided daily F_{CH_4} into growing season and fallow season periods to capture differences in processes. Models were developed across years for both the growing and fallow seasons and for each individual growing season (excluding 2009 due to the limited sample size). Individual fallow periods were not modeled due to the lack of flux coverage in winter; on average there were 153 measurements of daily F_{CH_4} during the growing season (excluding 2009) and only 58 measurements in the winter period.

First, we created multivariate linear regression models to investigate the relationships between environmental and flux drivers and daily F_{CH_4} . Daily F_{CH_4} was estimated by taking the mean of quality-controlled half-hourly fluxes for days with gaps of less than 8 h, and values were log transformed before being fit with linear models [Chu *et al.*, 2014; Treat *et al.*, 2007; Yan *et al.*, 2005]. We used a forward stepwise approach, where driving variables were ranked based on their pairwise correlation with F_{CH_4} and additional variables were included in the model if (1) there was a significant pairwise correlation, (2) their effect in the model was significant, (3) they improved the overall R^2 of the model, and (4) their inclusion resulted in an AIC reduction [Morin *et al.*, 2014]. The order in which the variables were added to successive multiple regression models was assessed by minimizing the AIC score of the model, and the model was expanded in a stepwise hierarchical manner to include variables until the addition of variables resulted in an increase in the AIC score, suggesting an overparameterized model.

Once we found the optimal linear model, a similar stepwise approach was used for the ANNs, where the variables from the linear model were used one by one and the incremental increase in R^2 and AIC values were calculated. To decrease processing time, rather than testing all potential variable orders, we included variables in the same order in the ANNs as in the linear models [Morin *et al.*, 2014]. Following the addition

of all variables identified as significant in the linear models, we also tried including those that were not identified as significant since neural networks can better capture nonlinear relationships. However, in the majority of cases if a variable was rejected from the linear models, it was also rejected from the ANNs, with the notable exception of WTD in neural network of fallow season F_{CH_4} . Slight modifications to the ANN were made to account for a decrease in sample size when using daily F_{CH_4} rather than half-hourly fluxes. Unlike the ANN gap-filling procedure described above, explanatory data were not divided into clusters and the fraction of the random samples going to the training, validation, and testing subsamples was 50%, 30%, and 20%, respectively. In addition, the complexity of the neural network architecture was decreased, ranging from one hidden layer with three nodes to two hidden layers with five and three nodes, respectively. In this case, for each added variable, we ran the ANN for one neural network architecture at a time (again with 20 extractions) and then chose the best architecture based on the calculated AIC value.

2.5.2. Diel Timescale

Due to their near-continuous nature, eddy covariance flux observations are also well suited for time series analysis [Hatala et al., 2012a; Koebsch et al., 2015; Novick et al., 2015; Stoy et al., 2005; Torrence and Compo, 1998]. We used the continuous wavelet transform with the Morlet mother wavelet to explore the correlation and time lag between F_{CH_4} and biophysical factors at the diel scale. Relative to Fourier analysis, wavelet transforms are much better suited for spectral analysis of nonstationary time series such as eddy covariance data [Katul et al., 1998; Stoy et al., 2005; Torrence and Compo, 1998]. The wavelet coherence spectrum represents the local correlation between two signals in time-frequency space from which a phase angle can be calculated and used to establish potential causal relationships [Grinsted et al., 2004]. The statistical significance of wavelet power was tested against the null hypothesis of a red noise first-order autoregressive process with Lag-1 autocorrelation [Grinsted et al., 2004]. For diel time periods with significant wavelet coherence, the phase angle was used to calculate the time lag between the two signals. To avoid spurious correlations between F_{CH_4} and driving variables, we did not use gap-filled data and instead replaced missing values with the median for the whole growing season [Hatala et al., 2012a], and time series were standardized to have a zero mean and unit variance [Hatala et al., 2012a; Stoy et al., 2005]. All analyses described above were performed using MATLAB (MathWorks Inc., 2015, version 8.6.0).

3. Results

3.1. Weather Conditions and Water Table Management

Due to the severe drought and record high temperatures experienced in California from 2012 to 2015 [Griffin and Anchukaitis, 2014], climatic variability was greater than expected, and there was an increasing trend in temperature over the course of the study (Table 2 and Figures 2a and 2b). Mean annual T_a was coolest in 2010–2011 (14.0°C), with largely negative deviations in mean monthly T_a , while 2014–2015 was the warmest year of the study (15.8°C), with predominantly positive deviations in T_a (Figure 2a). Soil temperatures showed a similar pattern as T_a (Figure 2b); however, the range in mean annual and seasonal T_s (>3.0°C) was larger than that of T_a (>1.8°C) (Table 2) since rice management (e.g., PAI and water regulation) in addition to T_a played a role in modulating T_s .

Water table depth at the paddy was actively managed throughout the study, and the drought did not result in water restrictions at the site. The field was typically flooded 45 to 60 days after planting and drained 30 to 45 days before harvest (Figure 2d). Therefore, over the course of the growing season there was standing water on the field between 64 and 112 days, with water levels generally maintained at 10 to 20 cm above the soil surface. In addition, the field was flooded for about 65 to 135 days during the fallow winter period.

3.2. CO₂ Fluxes

While the rice paddy was generally a net CO₂ sink during the growing season, large interannual differences were observed, with the site ranging from a large net CO₂ sink in the 2010 growing season (−607 g C-CO₂ m^{−2}) to near-CO₂ neutral in 2015 (8 g C-CO₂ m^{−2}) (Table 2 and Figure 3). Large interannual variability was also observed in growing season GEP and ER (Figure 3); GEP budgets differed by more than 500 g C-CO₂ m^{−2} yr^{−1} over the seven growing seasons, while cumulative growing season ER ranged from 782 to 1152 g C-CO₂ m^{−2} (Table 2). Interannual variability in growing season ER was strongly related to mean growing season T_s (Figure 5b). In addition, early drainage (i.e., a month earlier than the average drainage date) of the rice paddy in the 2015

Table 2. Annual, Growing, and Fallow Season Mean Air (T_a) and Soil Temperatures (T_s) at 2 cm Depth and Sums of Photosynthetically Active Radiation (PAR), Net CO_2 Flux (F_{CO_2}), Gross Ecosystem Production (GEP), Ecosystem Respiration (ER), CH_4 Flux (F_{CH_4}), and Annual Carbon (C) Budget^a

	Annual	Growing Season	Fallow Season
	T_a ($^{\circ}\text{C}$)		
2009–2010	15.0	19.6	9.7
2010–2011	14.0	18.6	9.0
2011–2012	14.0	18.7	10.7
2012–2013	14.7	19.0	8.8
2013–2014	15.4	20.2	11.6
2014–2015	15.8	20.5	12.0
	–	19.4	–
	T_s ($^{\circ}\text{C}$)		
2009–2010	14.6	18.4	10.1
2010–2011	15.1	18.7	11.0
2011–2012	15.0	19.6	11.3
2012–2013	15.8	19.8	11.1
2013–2014	16.5	20.9	13.3
2014–2015	17.6	22.0	13.7
2015–2016	–	21.4	–
	PAR (mol m^{-2})		
2009–2010	14,427	10,427	4,044
2010–2011	14,423	10,533	4,331
2011–2012	14,636	9,506	6,426
2012–2013	15,249	9,337	3,625
2013–2014	15,273	9,960	6,619
2014–2015	15,332	9,281	5,449
2015–2016	–	9,436	–
	F_{CO_2} ($\text{g C-CO}_2 \text{ m}^{-2}$)		
2009–2010	-167 ± 96	-486 ± 26	339 ± 91
2010–2011	-396 ± 46	-607 ± 19	264 ± 47
2011–2012	-33 ± 49	-261 ± 28	307 ± 39
2012–2013	32 ± 65	-294 ± 36	227 ± 49
2013–2014	46 ± 34	-335 ± 10	455 ± 33
2014–2015	316 ± 50	-192 ± 10	478 ± 51
2015–2016		8 ± 15	
	GEP ($\text{g C-CO}_2 \text{ m}^{-2}$)		
2009–2010	1340 ± 152	1268 ± 73	67 ± 134
2010–2011	1648 ± 74	1535 ± 59	97 ± 59
2011–2012	1328 ± 94	1182 ± 60	195 ± 78
2012–2013	1304 ± 130	1198 ± 85	69 ± 70
2013–2014	1555 ± 96	1431 ± 70	146 ± 66
2014–2015	1445 ± 107	1343 ± 86	89 ± 80
		1014 ± 75	
	ER ($\text{g C-CO}_2 \text{ m}^{-2}$)		
2009–2010	1173 ± 124	782 ± 29	406 ± 107
2010–2011	1252 ± 54	928 ± 32	362 ± 42
2011–2012	1361 ± 66	921 ± 21	502 ± 64
2012–2013	1336 ± 76	904 ± 39	296 ± 54
2013–2014	1601 ± 60	1095 ± 12	600 ± 55
2014–2015	1761 ± 60	1152 ± 18	567 ± 58
2015–2016	–	1022 ± 25	–
	F_{CH_4} ($\text{g C-CH}_4 \text{ m}^{-2}$)		
2009–2010	6.4 ± 4.3	2.5 ± 2.5	4.0 ± 2.0
2010–2011	12.1 ± 2.0	6.2 ± 0.7	7.0 ± 2.1
2011–2012	7.8 ± 1.0	4.0 ± 0.4	3.0 ± 1.0
2012–2013	6.7 ± 1.8	4.1 ± 0.3	2.1 ± 1.4
2013–2014	13.7 ± 2.0	8.9 ± 0.2	5.0 ± 2.0
2014–2015	19.3 ± 1.4	11.1 ± 0.2	8.1 ± 1.4
2015–2016	–	9.3 ± 0.4	–
	C Budget (g C m^{-2})		
2009–2010	5 ± 96	–	–
2010–2011	-232 ± 46	–	–
2011–2012	253 ± 49	–	–

Table 2. (continued)

	Annual	Growing Season	Fallow Season
2012–2013	237 ± 65	–	–
2013–2014	312 ± 34	–	–
2014–2015	524 ± 50	–	–
	–	–	–

^aThe annual carbon budget includes the amount of C lost as CH₄ and removed from the field through harvest. Error bounds indicate the 95% confidence interval for the gap-filling procedure.

growing season (Figure 2d), coupled with warm late-summer temperatures (Figure 2b), resulted in high rates of respiration and positive F_{CO_2} in the later part of the growing season (Figures 3b and 3c), contributing to the near-neutral net CO₂ budget that season.

When including the fallow season, the site ranged from being a strong net sink of CO₂ in 2010–2011 (−396 g C-CO₂ m^{−2} yr^{−1}) to a considerable net source in 2014–2015 (316 g C-CO₂ m^{−2} yr^{−1}) (Table 2). Annual F_{CO_2} was significantly related to annual ER ($p < 0.05$, $r^2 = 0.67$), but not GEP ($p = 0.43$, $r^2 = 0.16$), suggesting that interannual variability in F_{CO_2} at the paddy was driven more by changes in ER than GEP. Similar to growing season ER, interannual variability in annual ER was strongly related to mean annual T_s (Figure 5a), which explains why the rice paddy was a net CO₂ sink in the earlier, cooler years, and a net CO₂ source in the later, warmer years (Table 2).

3.3. Temporal Variability in CH₄ Fluxes and Annual Budgets

Daily F_{CH_4} increased throughout the growing season, reaching a maximum toward grain ripening, with a large pulse of CH₄ to the atmosphere following drainage of the field for harvest (Figure 3d). Although the overall seasonal pattern of F_{CH_4} was similar between years, the magnitude of the fluxes varied by an order of magnitude (Figure 3d). Lowest daily F_{CH_4} was observed in the 2009 summer period when, prior to drainage, emissions were <20 mg C-CH₄ m^{−2} d^{−1}. In contrast, daily F_{CH_4} during the growing season of 2014 exceeded 200 mg C-CH₄ m^{−2} d^{−1} prior to the drop in WTD for harvest. Emissions during the fallow season were generally low (~10 to 30 mg C-CH₄ m^{−2} d^{−1}) except when the field was drained for planting, which caused a large release of stored CH₄ (Figure 3d).

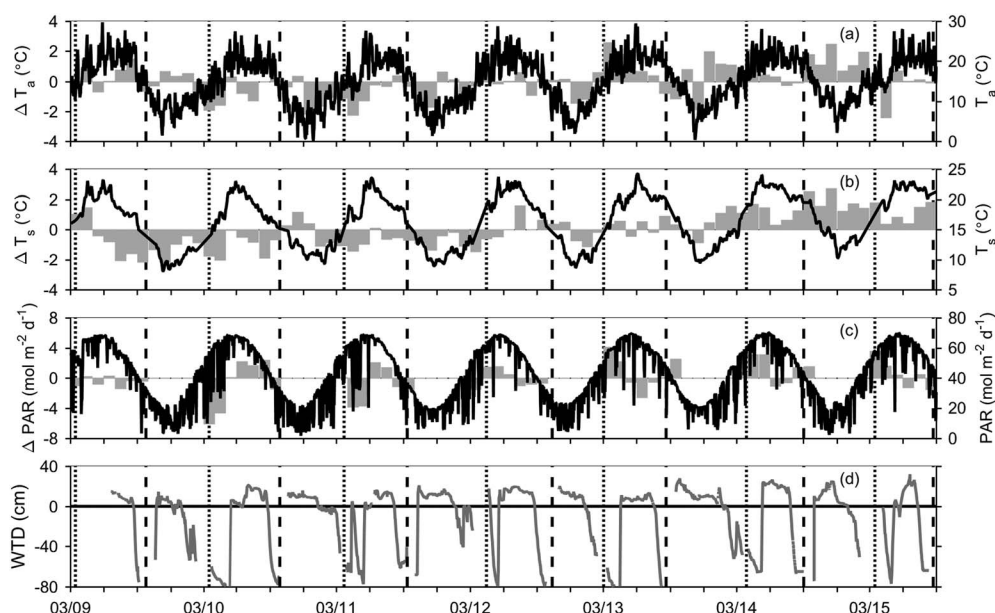


Figure 2. Monthly anomaly (grey bars) and time series of daily micrometeorological variables (solid lines), including (a) air temperature (T_a), (b) soil temperature at 32 cm depth (T_s), (c) photosynthetically active radiation (PAR), and (d) water table depth (WTD). Dotted and dashed lines indicate planting and harvest dates, respectively.

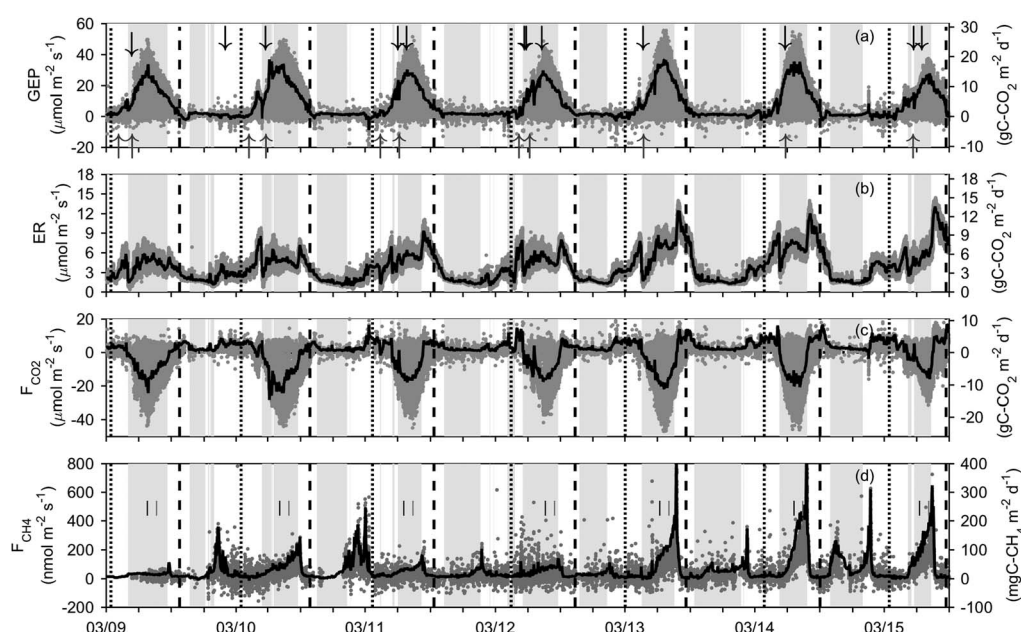


Figure 3. Times series of half-hourly (grey circles) and daily (solid lines) (a) gross ecosystem production (GEP), (b) ecosystem respiration (ER), (c) net ecosystem CO₂ exchange (F_{CO_2}), and (d) net ecosystem CH₄ exchange (F_{CH_4}). Half-hourly fluxes were not gap-filled, and daily data were integrated from gap-filled half-hourly data using the artificial neural network method. Shaded grey areas in the background of all panels indicate periods when the field was flooded. Dotted and dashed lines indicate planting and harvest dates, respectively. Herbicide application dates are indicated by the black downward pointing arrows, and fertilizer application dates are indicated by the upward facing grey arrows in Figure 3a. Panicle initiation and heading dates are indicated in Figure 3d by black and grey dashes, respectively.

Strong diel variation in F_{CH_4} was observed during the growing seasons of 2011 and 2013–2015, while weak or no consistent diel patterns were observed in other years (Figures 4 and S2–S4). In years with strong daily variation in F_{CH_4} , the diel peak in F_{CH_4} occurred in midafternoon between 14:00 and 15:30 h. Furthermore, in addition to large differences in the magnitude of the fluxes, the amplitude of the diel cycle varied strongly between years, increasing exponentially with mean growing season T_s (Figure 4h). No diel patterns were observed when the field was flooded during the winter months.

Annual CH₄ budgets varied considerably throughout the study, with over 3 times more CH₄ emitted in 2014–2015 than in 2009–2010 (Table 2). Growing season CH₄ emissions ranged from 2.5 to 9.3 g C-CH₄ m⁻² in 2009 and 2014, respectively, and wintertime F_{CH_4} was greatest in 2014–2015 (8.1 g C-CH₄ m⁻²) and considerably less CH₄ was released over the 2012–2013 fallow period (2.3 g C-CH₄ m⁻²).

3.4. CH₄ Emission Predictors

3.4.1. Interannual Scale

Interannual differences in annual and growing season CH₄ emissions were not significantly correlated with annual or growing season GEP ($p > 0.24$), F_{CO_2} ($p > 0.24$), or the number of drained or flooded days ($p > 0.14$). However, like ER (Figures 5a and 5b), annual and growing season F_{CH_4} were significantly correlated with mean annual or growing season T_s , respectively (Figures 5d and 5e). Over the full year, F_{CH_4} was most strongly related to T_s at intermediate depths (8 and 16 cm) since warmest T_s and presumably greatest CH₄ production were observed near the soil surface in the summer and deeper in the soil profile in winter, while during the growing season F_{CH_4} was most significantly correlated with T_s at 2 cm. As expected from these observations, there was a strong relationship between annual or growing season ER and F_{CH_4} ($p = 0.029$, $r^2 = 0.74$ and $p = 0.0011$, $r^2 = 0.90$, respectively). Increasing T_s also resulted in a significant increase in the annual or growing season CH₄ to GEP ratio (Figures 5g and 5h). Over the whole year, the ratio of F_{CH_4} (g C-CH₄ m⁻² yr⁻¹) to GEP (g C-CO₂ m⁻² yr⁻¹) ranged from 0.5% in 2009–2010 to 1.4% in 2014–2015 (Figure 5g), while during the growing season this ratio varied between 0.2% in 2009 and 0.9% in 2015 (Figure 5h). No variables were found that were significantly correlated with interannual differences in cumulative fallow season F_{CH_4} , including T_s (Figure 5f) or fallow season ER ($p = 0.42$).

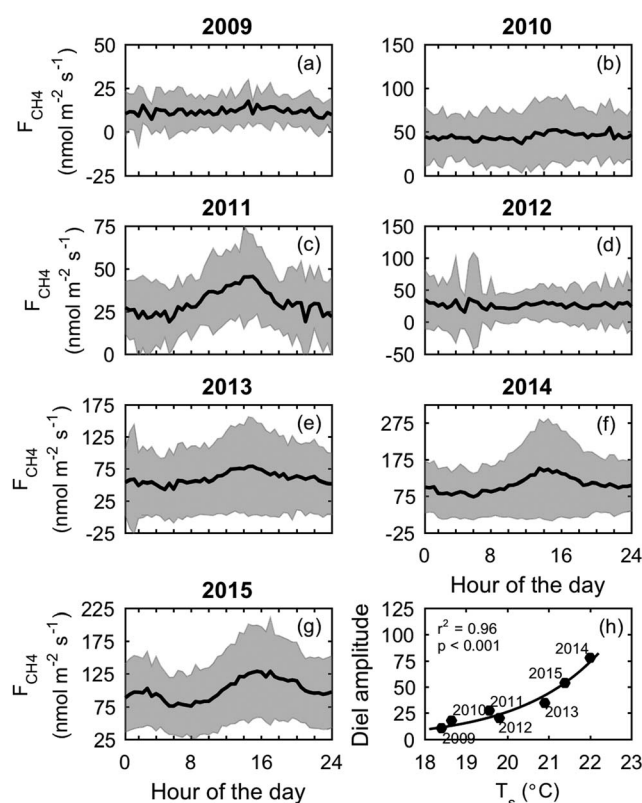


Figure 4. (a–g) Mean diel variation in CH_4 flux (F_{CH_4}) during the growing season when the field was flooded and (h) amplitude of the mean diel variation in F_{CH_4} as a function of mean growing season soil temperature (T_s) at 2 cm. The shaded area in Figures 4a–4g indicates the standard deviation. Note the difference in the vertical axes.

showed that regression coefficients were clustered into two groups, where slopes in 2009 to 2012 were significantly different from those in 2013 to 2015 (Figure 6). Regression coefficients appeared to be strongly influenced by mean growing season T_s , with smaller slopes observed in cooler years (2009 to 2012), and significantly larger slopes observed in warmer years (2013 to 2015) (Table 2), explaining the increasingly nonlinear relationship observed between daily GEP and F_{CH_4} in the latter years of the study (Figure 6). Within more than half the individual growing seasons, ANNs using just GEP and WTD as input variables were able to explain more than 90% of the variance in daily F_{CH_4} (Table 3), accurately capturing the nonlinear relationships associated with CH_4 emissions. However, across all growing seasons, GEP and WTD together only accounted for 46% of the variance in daily F_{CH_4} , indicating that additional variables are needed to account for the interannual variability in CH_4 emissions.

While mean growing season T_s was a significant factor explaining interannual differences in cumulative growing season F_{CH_4} (Figure 5e), across all growing seasons ER rather than temperature appeared as the third variable in the ANN model, strongly increasing model R^2 (Table 3). The overall positive correlation between ER and CH_4 emissions indicates that ER and F_{CH_4} , both products of microbial metabolism, were similarly affected by environmental regulators across years. The fact that ER improved the model beyond the contribution of other direct environmental drivers, including temperature, suggests that the common effects of these environmental drivers on soil respiration and F_{CH_4} are complex and nonlinear.

While ER strongly informed the model across all growing seasons, within individual years, there was frequently no significant relationship between ER or T_s and F_{CH_4} . Since rice paddies are managed ecosystems, the timing of planting and flooding had a strong influence on the relationship between T_s and F_{CH_4} , whereby planting in mid-April to mid-May followed by flooding a month or more later resulted in a decoupling between T_s and F_{CH_4} ; during the period of active rice growth daily F_{CH_4} increased while daily mean T_s typically

3.4.2. Seasonal Scale

Unsurprisingly, the neural network models consistently resulted in overall higher R^2 values than the multiple linear models (Table 3). Across all growing seasons and within individual growing seasons, GEP and WTD were typically selected as the top two variables in the models. Both daily GEP and WTD were positively correlated with daily F_{CH_4} but exhibited nonlinear relationships with daily F_{CH_4} . WTD acted primarily as an “on-off switch” for F_{CH_4} , with the exception of drainage of the field for harvest when large pulses of CH_4 were released to the atmosphere, comprising between ~20 and 26% of the total CH_4 released throughout the growing season (Figure 3d). During the flooded summer period (excluding the end-of-season CH_4 pulse), daily F_{CH_4} was strongly related to daily GEP (Figure 6). Analysis of covariance indicated that the parameters of the linear regression model of log transformed daily F_{CH_4} against daily GEP differed significantly between years, while pairwise comparisons of the slopes using Bonferroni critical values

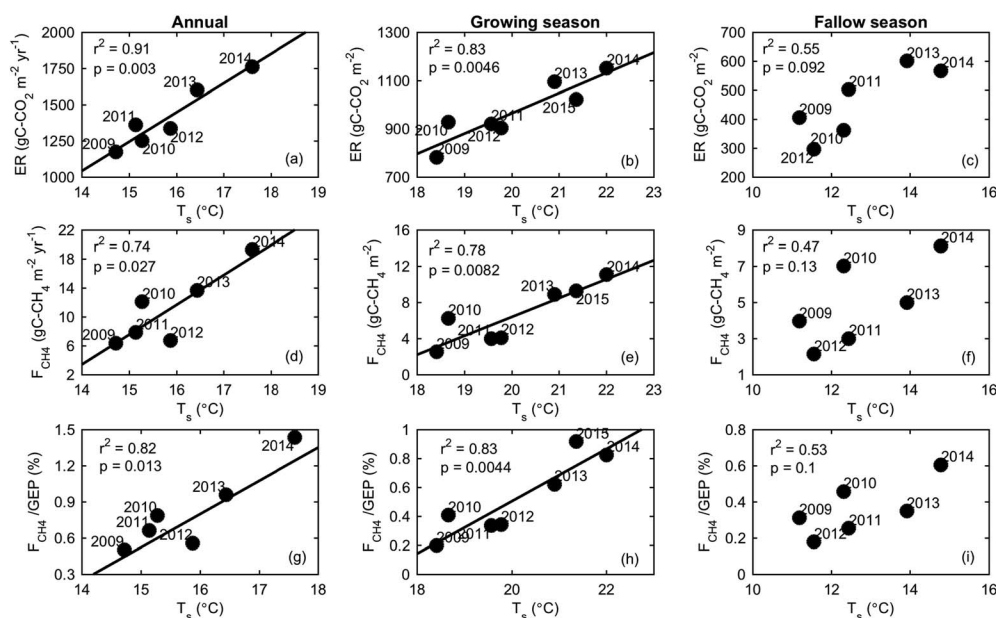


Figure 5. (a) Cumulative annual ecosystem respiration (ER) as a function of mean annual soil temperature (T_s) at 8 cm, (b) growing season ER as a function of mean growing season T_s at 2 cm, (c) fallow season ER as a function of mean fallow season T_s at 32 cm, (d) annual F_{CH_4} as a function of mean annual soil temperature (T_s) at 8 cm, (e) growing season F_{CH_4} as a function of mean growing season T_s at 2 cm, (f) fallow season F_{CH_4} as a function of mean fallow season T_s at 32 cm, (g) ratio of annual F_{CH_4} ($g\ C-CH_4\ m^{-2}\ yr^{-1}$) to GE ($g\ C-CO_2\ m^{-2}\ yr^{-1}$) as a function of mean annual T_s at 8 cm, (h) ratio of growing season F_{CH_4} ($g\ C-CH_4\ m^{-2}$) to GE ($g\ C-CO_2\ m^{-2}$) as a function of mean growing season T_s at 2 cm, and (i) ratio of fallow season of F_{CH_4} ($g\ C-CH_4\ m^{-2}$) to growing season GE ($g\ C-CO_2\ m^{-2}$) as a function of mean fallow season T_s at 32 cm.

decreased or remained near constant (Figures 2b and 3d). Consequently, even when there was a significant relationship between these variables, ER or T_s (or T_d) was typically only added as one of the later variables, resulting in only minor increases in model R^2 .

Within most years there was no significant relationship between u^* and F_{CH_4} , and across all years, u^* was only selected at the fifth variable in the model, explaining only minor variations in daily F_{CH_4} . LE was typically the third or fourth variable added to the models but generally also only resulted in relatively small increases in model R^2 .

During the fallow season, across all years (and within years; data not shown) daily F_{CH_4} was negatively correlated with T_s and ER (Table 3). Conversely, across all years there was a significant positive relationship between fallow season T_s and ER ($r = 0.75$, $p < 0.001$). The overall negative correlation between daily ER and F_{CH_4} during the fallow season suggests that during this period, the response of F_{CH_4} and ER to environmental drivers differed. LE was added as the second variable in the ANN model of fallow season daily F_{CH_4} , explaining an additional 21% of the variance in daily F_{CH_4} . Although WTD was not a significant predictor of fallow season F_{CH_4} in the stepwise linear model, the ANN was able to model the nonlinear interaction between daily F_{CH_4} and WTD, which is important for capturing the pulse of CH_4 to the atmosphere following water table drawdown in spring, as these CH_4 pulses accounted for on average $>50\%$ of the total CH_4 emitted during the fallow period.

3.4.3. Diel Scale

The diel pattern in F_{CH_4} from rice paddies has often been attributed to diel variation in T_s [Khalil et al., 1998; Meijide et al., 2011; Schütz et al., 1989a], which would be confirmed by the diel peak in T_s is temporally leading or coinciding with the peak in F_{CH_4} [Hatala et al., 2012a]. However, in all years with a pronounced and consistent diel pattern in growing season F_{CH_4} (2011 and 2013–2015), T_s lagged F_{CH_4} between 1.3 ± 0.04 h (2015) and 5.2 ± 0.06 h (2011) (Figures 7e, 7f, and 7i–7m). In contrast, the peak in GE led to that of F_{CH_4} with a mean time lag varying between 1.6 ± 0.03 h in 2011 and 4.4 ± 0.04 h in 2015 (Figures 7e, 7f, and 7i–7m). While the time series of both growing season T_s and F_{CH_4} and GE and F_{CH_4} were strongly coherent at the

Table 3. Statistical Tests (Correlation Coefficient (R), Coefficient of Determination (R^2), and Akaike Information Criterion (AIC)) for the Significance of Biophysical Drivers in the Models of Daily Methane Flux (F_{CH_4}), Including Pairwise Linear, Stepwise Multivariate Linear, and Hierarchical Neural Network Models^a

	Pairwise	Stepwise Linear		Neural Network	
	R	R^2	AIC	R^2	AIC
<i>Growing Season</i>					
2009–2015					
GEP	0.622	0.386	2853	0.258	7704
WTD	0.575	0.424	2793	0.462	7431
ER	0.168	0.458	2737	0.713	6853
LE	0.452	0.469	2718	0.751	6716
u^*	0.147	0.470	2718	0.765	6671
T_a	0.309	0.471	2717	0.814	6462
T_s	0.195	0.474	2712	0.825	6420
2015					
WTD	0.716	0.509	456	0.310	1248
GEP	0.620	0.671	402	0.930	931
T_s	0.307	0.682	398	0.943	914
LE	0.631	0.697	392	0.954	898
ER	-0.168	0.704	391	0.964	839
u^*	0.348	0.702	393 ^b	–	–
T_a	0.275	0.702	393 ^b	0.976	830
2014					
GEP	0.729	0.528	454	0.396	1321
WTD	0.704	0.557	445	0.942	981
LE	0.569	0.588	435	0.947	980
T_a	0.216	0.605	430	0.945	1004 ^b
u^*	0.059	–	–	–	–
T_s	-0.057	–	–	–	–
ER	-0.009	–	–	0.973	935
2013					
GEP	0.731	0.532	507	0.474	1322
ER	0.499	0.580	491	0.944	1000
WTD	0.493	0.594	486	0.957	968
LE	0.448	0.656	461	0.962	961
T_s	0.282	0.657	461	0.984	787
T_a	0.410	0.663	460	0.986	790 ^b
u^*	0.102	–	–	–	–
2012					
GEP	0.551	0.299	391	0.289	811
T_a	0.359	0.344	382	0.379	800
LE	0.238	0.348	382	0.725	709
WTD	0.469	0.399	371	0.731	691
T_s	0.319	0.397	373 ^b	–	–
ER	0.148	–	–	–	–
u^*	0.042	–	–	–	–
2011					
GEP	0.731	0.531	224	0.539	733
WTD	0.722	0.574	212	0.642	706
ER	0.296	0.649	185	0.781	642
T_a	0.413	0.656	183	0.792	642
LE	0.608	0.654	185 ^b	–	–
u^*	0.111	–	–	–	–
T_s	0.103	–	–	–	–
2010					
WTD	0.69	0.472	387	0.708	853
GEP	0.609	0.481	386	0.877	731
LE	0.466	0.488	385	0.905	720
T_a	0.318	0.496	384	0.910	735 ^b
ER	0.133	–	–	–	–
u^*	0.071	–	–	–	–
T_s	-0.07	–	–	–	–
<i>Fallow Season</i>					
2009–2014					

Table 3. (continued)

	Pairwise	Stepwise Linear		Neural Network	
	<i>R</i>	<i>R</i> ²	AIC	<i>R</i> ²	AIC
ER	-0.39	0.15	1079	0.089	2649
LE	0.237	0.201	1060	0.303	2566
<i>T</i> _s	-0.38	0.215	1055	0.444	2500
WTD	0.301	0.216	1056 ^b	0.617	2442
<i>T</i> _a	-0.345	0.216	1057 ^b	–	–
<i>u</i> [*]	0.062	–	–	–	–

^aDaily F_{CH_4} values were log transformed before being fit with linear models. Pairwise relationships indicate the direct (i.e., not incremental) correlation of each variable with daily F_{CH_4} . Note that italic *p* values indicate that the pairwise relationships were not significant and therefore not included in the stepwise linear models. Also, variables rejected from the stepwise linear models were tested in the neural network models and the results were only presented if the addition of the variable improved the overall R^2 of the model and was justified by a reduction in the AIC of the model.

^bIndicates that this variable was rejected from the model.

diel timescale (Figures S2 and S3), the time lag between GEP and F_{CH_4} in combination with phase-locked behavior (as indicated by the unimodal distribution of the phase angle; Figures 7f, 7j, 7l, and 7n) provides strong support that GEP and not T_s likely caused the diurnal pattern in F_{CH_4} . We also investigated the possibility that stomatal conductance (G_s) might drive F_{CH_4} , but as observed in other studies [e.g., Chanton *et al.*, 1997; Hatala *et al.*, 2012a] and consistent with the findings that stomata do not appear to be the major release site

of CH_4 from rice plants [Nouchi and Mariko, 1993], particularly in later growth stages [Wang *et al.*, 1997], there was no strong support for this mechanism (cf. supporting information).

Although gas transport in rice plants occurs primarily through molecular diffusion [Chanton *et al.*, 1997; Den van der Gon and Van Breemen, 1993], a minor component may be due to transpiration-induced bulk flow as suggested from isotopic evidence and peak diel F_{CH_4} coinciding with maximum daily transpiration rates [Chanton *et al.*, 1997]. In this study, LE and F_{CH_4} were also significantly coherent at the daily period (Figure S4) and varied nearly in phase (Figures 7f, 7j, 7l, and 7n), indicating that the diel variation in F_{CH_4} could also in part be associated with transpiration.

4. Discussion

4.1. Interannual Variability in F_{CH_4}

Soil temperature, which experienced large variability throughout the study due to record high temperatures experienced during this period [Griffin and Anchukaitis, 2014], explained much of the interannual variability in annual and growing season ER and F_{CH_4} . This likely results from increased microbial

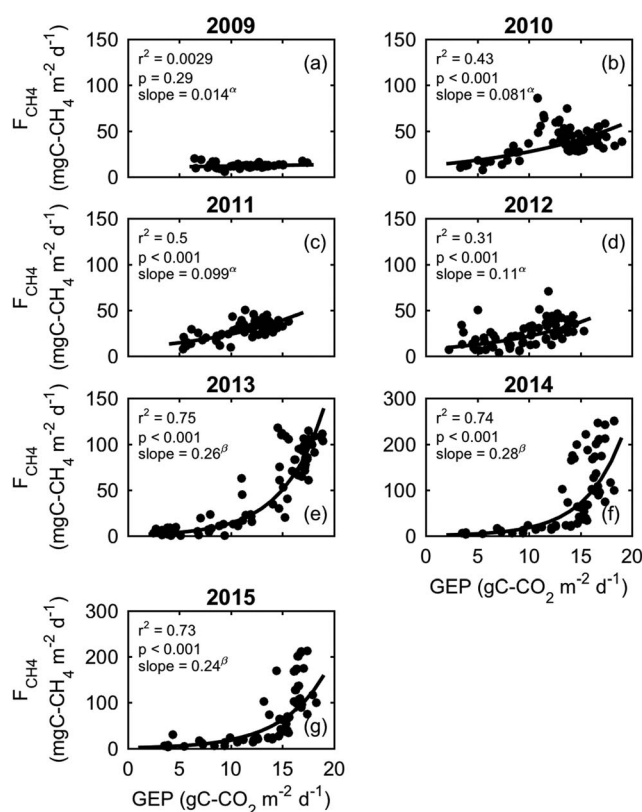


Figure 6. Linear regression models of log transformed daily CH_4 emission (F_{CH_4}) against daily gross photosynthesis (GEP) during the growing season from the time when the field was flooded to when water levels began to drop in preparation for harvest. Separate model parameters are fitted for each year. The lack of a significant relationship in 2009 is likely due to the larger amount of data that were excluded in that year (cf. section 2). Greek letters indicate statistically separate regression coefficients (i.e., slopes). Note the difference in the vertical axis in Figures 6f and 6g.

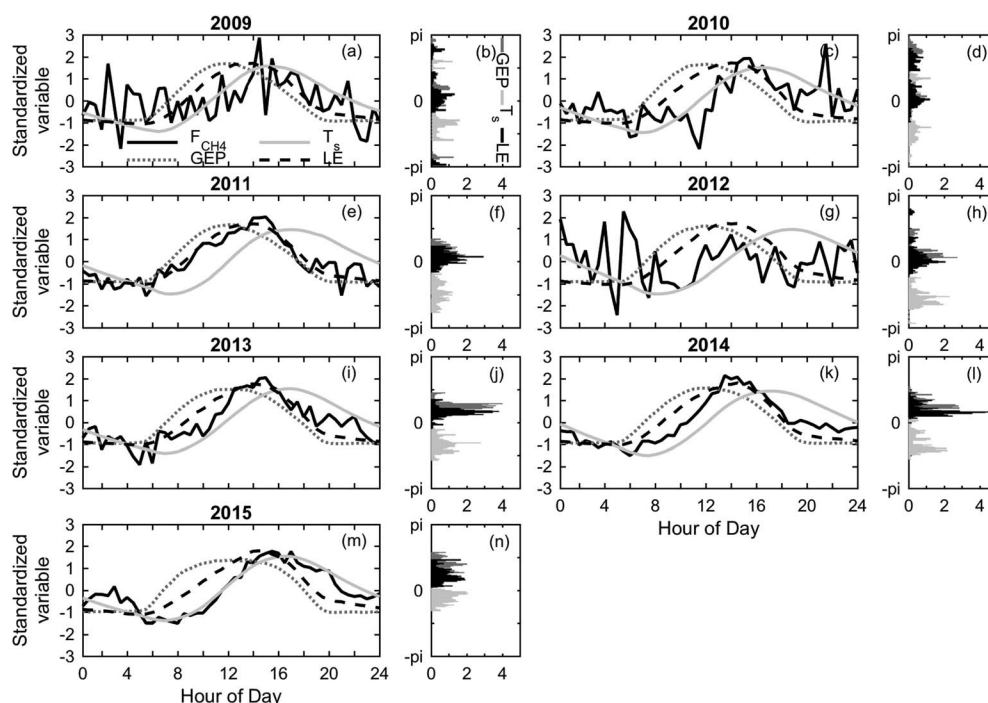


Figure 7. (a, c, e, g, i, k, and m) Standardized (i.e., zero mean and unit variance) mean diurnal variation in F_{CH_4} (black line), photosynthesis (GEP) (dotted line), soil temperature (T_s) at 2 cm (gray line), and latent heat flux (LE) (dashed line) during the growing season averaged over the same period as Figure 4 and (b, d, f, h, j, l, and n) distribution of the phase angles between F_{CH_4} and GEP (dark gray bars), T_s (light gray bars), or LE (black bars) at the daily timescale for periods when there was significant coherence.

activity with higher soil temperatures [Conrad, 2007; Yvon-Durocher et al., 2014] and in the case of F_{CH_4} , stronger temperature effects on CH_4 production than consumption, which is consistent with greater temperature sensitivity of methanogenesis than methanotrophy [Le Mer and Roger, 2001]. This is also supported by the positive correlation between annual or growing season ER and F_{CH_4} ; increased overall microbial activity (i.e., enhanced ER) resulted in greater CH_4 emissions, suggesting that higher temperatures disproportionately influenced CH_4 production over CH_4 consumption [Morin et al., 2014].

While several studies found that wetland net ecosystem production (NEP, where $NEP = -F_{CO_2}$) or plant biomass was positively correlated with F_{CH_4} [Bellisario et al., 1999; Chanton et al., 1997; Huang et al., 1997b; Sass et al., 1990; Whiting and Chanton, 1993], in this study, the growing season F_{CH_4} to GEP ratio varied considerably from year to year, and GEP was not a significant predictor of F_{CH_4} . However, the ratio of annual or growing season F_{CH_4} to GEP was significantly correlated with mean T_s , which is consistent and expected from the considerably higher temperature dependence of ecosystem level CH_4 emissions relative to photosynthesis [Yvon-Durocher et al., 2014], and follows the findings of Yvon-Durocher et al. [2014] who observed an increase in the CH_4 to CO_2 ratio with seasonal increases in temperature across a range of wetland sites. Our observation of an increase in the ratio of CH_4 to GEP with increasing temperature has implications for upscaling F_{CH_4} from both rice paddies and wetland ecosystems.

Using the relationship between F_{CH_4} and rice productivity, several studies have estimated CH_4 emissions from rice paddies at regional to global scales assuming that a constant fraction of net primary productivity (NPP) is released as CH_4 [Aselmann and Crutzen, 1989; Bachelet et al., 1995; Bachelet and Neue, 1993; Taylor et al., 1991]. However, given the large interannual variability in the ratio of F_{CH_4} to photosynthesis, upscaling CH_4 fluxes by estimating emissions as a constant fraction of NPP ignores an important source of variability [Cao et al., 1995; Denier van der Gon et al., 2000]. Our results suggest that instead of estimating F_{CH_4} as a constant fraction of productivity, F_{CH_4} can be estimated as a variable fraction of GEP, which changes linearly with T_s . This relationship can potentially be used to help improve global estimates of CH_4 emissions from rice paddies. However, whether this relationship holds across or within other sites remains to be tested since few

Table 4. Growing Season Mean Air Temperature (T_a), Net CO₂ Flux (F_{CO_2}), CH₄ Emissions (F_{CH_4}), and Ratio of CH₄ Flux (F_{CH_4}) to GEP From the Literature as Compared to the Values Reported in This Study

Location	Growing Season	Mean T_a^c (°C)	F_{CO_2} (g C-CO ₂ m ⁻²)	F_{CH_4} (g C-CH ₄ m ⁻²)	F_{CH_4} /GEP (%)	Reference
California, USA	2009	19.6	-486	2.5	0.2	This study
California, USA	2010	18.6	-607	6.2	0.4	This study
California, USA	2011	18.7	-261	4.0	0.3	This study
California, USA	2012	19.0	-294	4.1	0.3	This study
California, USA	2013	20.2	-335	8.9	0.6	This study
California, USA	2014	20.5	-192	11.1	0.8	This study
California, USA	2015	19.4	8	9.3	0.9	This study
	Average	19.4	-310	6.6	0.5	This study
Laguna, Philippines	2008 DS ^a	27.2	-241			Alberto et al. [2012]
Laguna, Philippines	2008 WS ^b	27.7	-375			Alberto et al. [2012]
Laguna, Philippines	2009 DS	27.2	-402			Alberto et al. [2012]
Laguna, Philippines	2009 WS	27.7	-293			Alberto et al. [2012]
Laguna, Philippines	2010 DS	27.2	-311			Alberto et al. [2012]
Odisha, India	2012 WS	31	-355	8.4	1.3	Bhattacharyya et al. [2014]
Odisha, India	2012–2013 DS	23.4	-341	12.9	2	Bhattacharyya et al. [2014]
Ibaraki, Japan	2002	20.7	-400			Saito et al. [2005]
Ibaraki, Japan	2003	23	-354	9.3	1.3	Miyata et al. [2005]
Okayama, Japan	2003	20	-438			Miyata et al. [2005]
Texas, USA	1994 and 1995	25.2			1.4 ^d	Huang et al. [1997a]
California, USA	2002		-789			McMillan et al. [2007]

^aDS = dry season.^bWS = wet season.^cNote that air temperature (T_a) was reported instead of soil temperature (T_s) since most studies did not report T_s .^dBased on the mean carbon ratio of CH₄ to NPP of 2.8% and assuming NPP = 0.5GPP.

studies have made simultaneous measurements of CO₂ and CH₄ fluxes from rice paddies over the course of a full growing season or year. Based on a limited number of studies, the ratios of F_{CH_4} to GEP reported for other rice paddies are slightly higher than the values measured in this study, which is consistent with warmer temperatures at these locations (Table 4). The F_{CH_4} to GEP ratio may also be influenced by other factors known to influence CH₄ emissions. For instance, Huang et al. [1997a] found that the carbon ratio of seasonal CH₄ emission to NPP (deduced from grain yield) varied from 1.2% to 5.4%, with an average value of 2.8%, and that this ratio was soil and variety dependent. Additional variables that could influence this correlation include fertilizer type and application rate [Cicerone et al., 1992; Lindau et al., 1991; Naser et al., 2007; Schütz et al., 1989a; Yagi and Minami, 1990; Yan et al., 2005], grain to biomass ratio [Denier van der Gon et al., 2002], and floodwater management [Alberto et al., 2014; Linquist et al., 2015; Wassmann et al., 2000a; Yagi et al., 1996].

Cumulative growing season F_{CH_4} , which ranged from 2.5 to 11.1 g C m⁻² (or 13 to 71 mg C-CH₄ m⁻² d⁻¹), was low compared to other rice systems [cf. IPCC, 1997, Tables 4–9; Yan et al., 2005]. For continuously flooded rice without organic amendment, CH₄ emissions are estimated to average 135 mg C-CH₄ m⁻² d⁻¹ [Yan et al., 2005]. Low CH₄ production in some peatlands has been attributed the fact that these soils have large amounts of high molecular weight organic compounds that can be very resistant to microbial degradation [Bridgham et al., 2013], which is in agreement with a recent study at our site that showed that the highly organic soil contained a large abundance of lignin and other aromatics, lipid, and aliphatics, which are likely not readily decomposable due to their chemical recalcitrance under anoxic conditions and can also be highly inhibitory to soil microorganisms [Ye et al., 2016]. Cooler soil temperatures at our site (typically in the range of 4 ± 2°C cooler; cf. Table 2 [Huang et al., 1998]) compared to tropical or subtropical sites may also explain lower CH₄ fluxes [Neue and Sass, 1994; Wassmann et al., 2000b]. Additional factors potentially contributing to low F_{CH_4} at our site may be differences in rice cultivar [Ding et al., 1999; Huang et al., 1997a; Wassmann and Aulakh, 2000], soil properties [Aulakh et al., 2001; Bachelet and Neue, 1993; Sass et al., 1994] including the presence of electron acceptors, particularly ferric iron at this site [Ye et al., 2015], and the relatively recent conversion of this site to rice agriculture which influences the amount of labile soil organic C, redox dynamics, and microbial community changes [Eusufzai et al., 2010]. Seasonal CH₄ totals lower than the ones reported in this study are generally observed in rice paddies with little or no organic amendments and/or midseason or intermittent drainage [e.g., Alberto et al., 2014; Bossio et al., 1999; Bronson et al., 1997; Cicerone et al., 1992;

Ding et al., 1999; Eusufzai et al., 2010; Linqvist et al., 2015; Nishimura et al., 2004; Yagi and Minami, 1990; Yagi et al., 1996]. Despite high C content soils and prolonged flooding periods, rice in the Delta emits less CH_4 relative to rice crops in warmer climates while also reducing the loss of soil through peat oxidation.

4.2. Seasonal Variability in F_{CH_4}

While daily F_{CH_4} in natural wetlands has been found to be strongly explained by T_s [Chu et al., 2014; Herbst et al., 2013; Sturtevant et al., 2015], the lack of a relationship between growing season F_{CH_4} and T_s is commonly reported in the literature [Cicerone et al., 1983; Huang et al., 1997b, 1998; Neue and Sass, 1994; Sass et al., 1990; Schütz et al., 1990], since the timing of management decisions can mask the relationship between F_{CH_4} and soil temperature. Instead, the seasonal variability in F_{CH_4} may be due to the influence of other overriding factors such as plant growth and development [Le Mer and Roger, 2001; Sass et al., 2000]. The relationship between plant productivity and F_{CH_4} is well established [Chanton et al., 1997; Hatala et al., 2012b; Huang et al., 1997b; Whiting and Chanton, 1993], and similarly, we observed a strong relationship between daily F_{CH_4} and GEP throughout the growing season, with GEP being an important predictor of daily F_{CH_4} . Increased productivity can enhance CH_4 emissions by providing a greater number and larger diameter conduits from the reducing soil to the atmosphere and by providing C substrates for CH_4 production [Chanton et al., 1997; Le Mer and Roger, 2001]. Another possible effect of increased plant activity would be an increased supply of oxygen to the rhizosphere stimulating CH_4 oxidation; however, the results from this study suggest a net stimulation of CH_4 production by rice plants. Although variations in F_{CH_4} during the crop cycle were not attributed to temperature variations, T_s did appear to have a strong influence on the amount of fixed C respired as CH_4 , as indicated from the significantly larger slopes and increasingly nonlinear relationship between daily GEP and F_{CH_4} in the warmer, latter years of the study.

Water table depth was also a major controlling factor of seasonal F_{CH_4} , both during the growing season and fallow period. WTD plays a key role in regulating the balance between CH_4 production and oxidation by influencing the depth of anaerobic and aerobic zones in the soil [Whalen, 2005]. Fluctuations in water levels were also associated with large pulses of CH_4 to the atmosphere. As expected, ANNs better captured the nonlinear influence of WTD on F_{CH_4} .

Friction velocity and LE had considerably less influence on daily F_{CH_4} than GEP and WTD over the course of the growing season. Increased turbulent mixing (i.e., higher u^*) can enhance CH_4 transport, although turbulent mixing has typically been found to be more important in regulating shorter-term (e.g., hourly) CH_4 fluxes [Chu et al., 2014; Koebisch et al., 2015]. LE has been observed to be an important driver of F_{CH_4} in wetlands across multiple timescales, as increased evaporation can enhance convective mixing at the water surface enhancing CH_4 transport, and CH_4 exchange and transpiration are linked through the mechanism of internal gas transport in wetland plants [Morin et al., 2014; Sturtevant et al., 2015]. However, the weaker dependence of daily F_{CH_4} on LE observed in this study can be explained by the fact that transpiration likely dominated LE for most of the growing season and that plant-mediated gas transport in rice plants appears to be independent [Neue et al., 1997; Seiler et al., 1983] or only weakly dependent on transpiration rates [Chanton et al., 1997]. However, during the fallow period, LE was an important driver of F_{CH_4} , potentially reflecting enhanced convective mixing and CH_4 transport with increased evaporation.

As observed by Morin et al. [2014], ER was an important variable for explaining CH_4 emissions. Across growing seasons, ER likely represented the end result of complex nonlinear processes that similarly influenced CH_4 exchange, in particular the influence of T_s since it was a significant driver of year-to-year differences in CH_4 emissions. Incorporating ER in gap-filling models for CH_4 can therefore inform more about CH_4 fluxes than models using only direct drivers [Morin et al., 2014]. While there was a positive relationship between ER and F_{CH_4} across growing seasons, a negative relationship was observed across fallow seasons. This negative relationship likely reflects differences in the transport pathways between CO_2 and CH_4 during this time, due in large part to solubility differences. CO_2 is highly soluble in water, and many lakes (similar to our flooded and fallow rice field) are oversaturated with CO_2 , and therefore, there is a high rate of diffusive loss of CO_2 at the air-water interface, particularly in the winter when productivity is low [Casper et al., 2000]. Conversely, CH_4 is much less soluble, with lower concentrations predicted in the surface water as a consequence of oxidation at the oxic/anoxic boundary, and therefore, CH_4 is more transport limited, with ebullition expected to be the dominant pathway of CH_4 emission to the atmosphere [Casper et al., 2000]. As a result of

these differences in transport pathways, ER and T_s were positively correlated, while F_{CH_4} was negatively correlated with ER and T_s during the fallow season.

4.3. Diel Variability in F_{CH_4}

Understanding drivers of diel variations in F_{CH_4} has important implications for improving mechanistic models of CH_4 flux and for daily and seasonal extrapolation of studies that only measure daytime F_{CH_4} [Hatala et al., 2012a]. While some studies report a diel pattern in CH_4 emissions from rice paddies [e.g., Bronson et al., 1997; Chanton et al., 1997; Ding et al., 1999; Satpathy et al., 1997; Schütz et al., 1989a; Yagi and Minami, 1993], others found none [Cicerone et al., 1983; Yagi and Minami, 1990]. Diel variations have often attributed to daily fluctuations in temperature [Den van der Gon and Van Breemen, 1993; Holzapfel-Pschorn and Seiler, 1986; Khalil et al., 1998]; however, several studies found no or poor correlations among these variables [Aulakh et al., 2001]. Using wavelet coherence [Grinsted et al., 2004; Torrence and Compo, 1998], we showed that the diurnal pattern in F_{CH_4} can largely be attributed to GEP and not T_s , consistent with an earlier study at this site that reported on the diel pattern for the 2011 growing season [Hatala et al., 2012a]. Additional support for this finding comes from the fact that no diel fluctuation in F_{CH_4} was observed during the fallow season although T_s showed a distinct diel variation, and from stable isotope labeling studies that showed a strong transient link between rice photosynthesis and F_{CH_4} , with time lags between CO_2 assimilation and CH_4 emission from about 2 h to 3 days [Dannenberg and Conrad, 1999; Minoda and Kimura, 1994; Minoda et al., 1996]. Although GEP appeared to induce the diel pattern in F_{CH_4} , we found that mean T_s significantly influenced the amplitude of diel CH_4 fluctuations, again suggesting that the amount of photosynthetic C respired as CH_4 is strongly dependent on T_s . Temperature can also influence CH_4 transport through rice plants with conductance for CH_4 increasing with temperature [Le Mer and Roger, 2001]. The increase in the amplitude of diel CH_4 variations with temperature also helps explain why diel patterns were observed in some years but not others.

Our results also suggest that in addition to photosynthesis, diel variations in F_{CH_4} may in part be linked with transpiration, as a minor component of gas transport in rice may be due to transpiration-induced bulk flow [Chanton et al., 1997]. However, since diel patterns were only observed in some years, while transpiration-induced bulk flow is expected to occur in all years, it is unclear whether daily fluctuations are linked with transpiration. Furthermore, other studies have found that plant-mediated gas transport is independent of transpiration rate [Neue et al., 1997; Seiler et al., 1983]. Similar to previous studies, we also ruled out the role of stomatal conductance in driving diel F_{CH_4} [Chanton et al., 1997; Hatala et al., 2012a; Nouchi and Mariko, 1993] since F_{CH_4} appears to be independent of stomatal opening as CH_4 is released to the atmosphere from various parts of the rice plant including from the culm and both the abaxial epidermis of the lower portion of leaf sheath and near the junction of the nodal region and leaf sheath [Nouchi and Mariko, 1993], with the development of nodes providing the major release pathway of CH_4 to the atmosphere at later growth stages [Wang et al., 1997]. Although this study strongly suggests that photosynthesis primarily caused the diel variation in F_{CH_4} , more studies are needed to explicitly understand the role of temperature, photosynthesis, transpiration, and other factors (e.g., methane oxidation and transport, pH, and microbial community changes) [Satpathy et al., 1997; Schütz et al., 1989b] in driving diel CH_4 patterns.

4.4. Interannual Variability in Net C Budgets

By monitoring the rice field for seven growing seasons, we observed large variability in seasonal and annual net CO_2 and C budgets. Net CO_2 uptake at the site generally fell within the range of values reported for other irrigated rice paddies around the world, with the exception of 2014 and 2015, when uptake was reduced or the site was near- CO_2 neutral, respectively (Table 4). Considering the full annual C budget, including the amount of C lost as CH_4 and removed from the field through harvest, the site was on average a net C source but showed considerable interannual variability with values ranging between -232 and $524 \text{ g C m}^{-2} \text{ yr}^{-1}$ (Table 2). Variability in net CO_2 budgets was strongly linked to the variability in ER, which is perhaps not unexpected as ER is a year-round process [Strachan et al., 2015], with higher ER and a shift toward a positive C balance under warmer conditions. This is similar to studies of temperate freshwater marshes where interannual climatic variability caused marsh C budgets to vary considerably among years as a result of warmer temperatures enhancing ER and F_{CH_4} [Chu et al., 2015; Strachan et al., 2015]. As observed in natural and restored wetlands [e.g., Helfter et al., 2015; Herbst et al., 2013; Rocha and Goulden, 2008; Roulet et al., 2007],

we show that wetlands converted to agricultural land, a widespread practice due to the economic benefit of the fertile soil [Kramer and Shabman, 1993], can exhibit strong interannual variability in CO₂ and CH₄ budgets. This highlights the need for more long-term flux measurements from both natural and managed wetlands under a range of climate conditions, since a warming climate is likely to have a strong influence on the net greenhouse gas balance and radiative forcing of these complex ecosystems [Petrescu et al., 2015].

5. Conclusion

In this paper, we present 6.5 years of eddy covariance measurements of CO₂ and CH₄ exchange from an irrigated rice paddy located in Northern California and investigate the factors affecting CH₄ emissions across diel to interannual scales by using time series analysis and empirical modeling. Improved understanding of the environmental drivers of CH₄ emissions is critical for improving our ability to adequately model whole-ecosystem CH₄ emissions [Bridgman et al., 2013]. Based on wavelet analysis, we showed that photosynthesis and not soil temperature was the dominant factor influencing the diel pattern in CH₄ flux, although soil temperature significantly influenced the amplitude of diel CH₄ fluctuations. Linear and neural network models indicated that during the growing season, photosynthesis and water levels explained much of the variance in daily average F_{CH_4} . This differs from natural wetlands where soil temperature has frequently been observed to be an important factor regulating seasonal CH₄ fluxes due to the strong impact of the timing of rice planting and flooding on relationships with temperature. However, temperature strongly influenced the amount of photosynthetic carbon respired as CH₄. LE and u^* generally only explained minor variations in daily F_{CH_4} during the growing season. Unsurprisingly, ANNs explained much more of the variance in daily F_{CH_4} than the linear models. ANNs were able to capture between 73% and 98% of the variance in daily growing season CH₄ exchange and 62% of the variance in fallow season fluxes. Differences in R^2 values between years and the percentage of unexplained variance may be attributable to management, as no management variables were included in the models. Future work could focus on explicitly incorporating management variables (e.g., fertilizer and herbicide application rates and timing, planting density, cultivar type, and straw residue) into empirical models of F_{CH_4} .

Soil temperature was a significant driver of interannual differences in annual and growing season CH₄ sums. Temperature also had a strong influence on the annual and growing season ratio of CH₄ flux to productivity, with this ratio increasing significantly with mean soil temperature. While previous studies have upscaled CH₄ emissions from rice by simply relating them to primary production, our results suggest that the relationship between the carbon ratio of CH₄ to photosynthesis and soil temperature may help improve global estimates of CH₄ flux from rice paddies. More studies, however, are needed to assess if this relationship holds across or within other rice paddies and wetlands in general. The pronounced variation in temperature observed throughout the study also had a strong influence on ecosystem respiration, which resulted in large interannual variability in net carbon budgets at the paddy, highlighting the need for long-term measurements particularly under changing climatic conditions.

Acknowledgments

The authors extend many thanks to Bryan Brock and the California Department of Water Resources for project funding (contract 4600008849). Additional funding was provided by the National Science Foundation (grant EF-1241960) and the U. S. Department of Agriculture (NIFA grant 2011-67003-30371), and funding for this AmeriFlux core site was provided by the U.S. Department of Energy's Office of Science (AmeriFlux contract 7079856). Data from the Twitchell Island rice paddy (US-Twt) are freely available from the AmeriFlux database (<http://ameriflux.lbl.gov/>). Additionally, funding for the AmeriFlux Management Project is provided by the U.S. Department of Energy's Office of Science under contract DE-AC02-05CH11231. S.H. Knox thanks Gavin McNicol for his valuable discussions in the preparation of this manuscript and Iryna Dronova for her assistance with the preparation of Figure 1. We also thank Gil Bohrer and another anonymous reviewer for their helpful comments to improve the manuscript.

References

- Alberto, M. C. R., T. Hirano, A. Miyata, R. Wassmann, A. Kumar, A. Padre, and M. Amante (2012), Influence of climate variability on seasonal and interannual variations of ecosystem CO₂ exchange in flooded and non-flooded rice fields in the Philippines, *Field Crops Res.*, 134, 80–94, doi:10.1016/j.fcr.2012.05.002.
- Alberto, M. C. R., R. Wassmann, R. J. Buresh, J. R. Quilty, T. Q. Correa Jr., J. M. Sandro, and C. A. R. Centeno (2014), Measuring methane flux from irrigated rice fields by eddy covariance method using open-path gas analyzer, *Field Crops Res.*, 160, 12–21, doi:10.1016/j.fcr.2014.02.008.
- Aselmann, I., and P. J. Crutzen (1989), Global distribution of natural freshwater wetlands and rice paddies, their net primary productivity, seasonality and possible methane emissions, *J. Atmos. Chem.*, 8(4), 307–358, doi:10.1007/BF00052709.
- Aubinet, M., et al. (1999), Estimates of the annual net carbon and water exchange of forests: The EUROFLUX methodology, in *Advances in Ecological Research*, edited by A. H. Fitter and D. G. Raffaelli, pp. 113–175, Academic Press, London, doi:10.1016/S0065-2504(08)60018-5.
- Aulakh, M. S., R. Wassmann, and H. Rennenberg (2001), Methane emissions from rice fields—Quantification, mechanisms, role of management, and mitigation options, in *Advances in Agronomy*, pp. 193–260, Academic Press, doi:10.1016/S0065-2113(01)70006-5.
- Bachelet, D., and H. U. Neue (1993), Methane emissions from wetland rice areas of Asia, *Chemosphere*, 26(1–4), 219–237, doi:10.1016/0045-6535(93)90423-3.
- Bachelet, D., J. Kern, and M. Tölg (1995), Balancing the rice carbon budget in China using spatially-distributed data, *Ecol. Modell.*, 79(1–3), 167–177, doi:10.1016/0304-3800(94)00031-C.
- Baldocchi, D. D. (2003), Assessing the eddy covariance technique for evaluating carbon dioxide exchange rates of ecosystems: Past, present and future, *Global Change Biol.*, 9(4), 479–492, doi:10.1046/j.1365-2486.2003.00629.x.

- Baldocchi, D. D. (2014), Measuring fluxes of trace gases and energy between ecosystems and the atmosphere—The state and future of the eddy covariance method, *Global Change Biol.*, 20(12), 3600–3609, doi:10.1111/gcb.12649.
- Baldocchi, D. D., and C. Sturtevant (2015), Does day and night sampling reduce spurious correlation between canopy photosynthesis and ecosystem respiration?, *Agric. For. Meteorol.*, 207, 117–126, doi:10.1016/j.agrformet.2015.03.010.
- Baldocchi, D. D., B. B. Hicks, and T. P. Meyers (1988), Measuring biosphere-atmosphere exchanges of biologically related gases with micrometeorological methods, *Ecology*, 69(5), 1331–1340, doi:10.2307/1941631.
- Baldocchi, D. D., M. Detto, O. Sonnentag, J. Verfaillie, Y. A. Teh, W. Silver, and N. M. Kelly (2012), The challenges of measuring methane fluxes and concentrations over a peatland pasture, *Agric. For. Meteorol.*, 153, 177–187, doi:10.1016/j.agrformet.2011.04.013.
- Bellisario, L. M., J. L. Bubier, T. R. Moore, and J. P. Chanton (1999), Controls on CH₄ emissions from a northern peatland, *Global Biogeochem. Cycles*, 13(1), 81–91, doi:10.1029/1998GB900021.
- Bhattacharyya, P., S. Neogi, K. S. Roy, P. K. Dash, A. K. Nayak, and T. Mohapatra (2014), Tropical low land rice ecosystem is a net carbon sink, *Agric. Ecosyst. Environ.*, 189, 127–135, doi:10.1016/j.agee.2014.03.013.
- Bossio, D. A., W. R. Horwath, R. G. Mutters, and C. van Kessel (1999), Methane pool and flux dynamics in a rice field following straw incorporation, *Soil Biol. Biochem.*, 31(9), 1313–1322, doi:10.1016/S0038-0717(99)00050-4.
- Bridgman, S. D., H. Cadillo-Quiroz, J. K. Keller, and Q. Zhuang (2013), Methane emissions from wetlands: Biogeochemical, microbial, and modeling perspectives from local to global scales, *Global Change Biol.*, 19(5), 1325–1346, doi:10.1111/gcb.12131.
- Bronson, K. F., H.-U. Neue, E. B. Abao, and U. Singh (1997), Automated chamber measurements of methane and nitrous oxide flux in a flooded rice soil: I. Residue, nitrogen, and water management, *Soil Sci. Soc. Am. J.*, 61(3), 981–987, doi:10.2136/sssaj1997.03615995006100030038x.
- Cao, M., J. B. Dent, and O. W. Heal (1995), Modeling methane emissions from rice paddies, *Global Biogeochem. Cycles*, 9(2), 183–195, doi:10.1029/94GB03231.
- Casper, P., S. C. Maberly, G. H. Hall, and B. J. Finlay (2000), Fluxes of methane and carbon dioxide from a small productive lake to the atmosphere, *Biogeochemistry*, 49(1), 1–19, doi:10.1023/A:1006269900174.
- Chanton, J. P., G. J. Whiting, N. E. Blair, C. W. Lindau, and P. K. Bollich (1997), Methane emission from rice: Stable isotopes, diurnal variations, and CO₂ exchange, *Global Biogeochem. Cycles*, 11(1), 15–27, doi:10.1029/96GB03761.
- Chu, H., J. Chen, J. F. Gottgens, Z. Ouyang, R. John, K. Czajkowski, and R. Becker (2014), Net ecosystem methane and carbon dioxide exchanges in a Lake Erie coastal marsh and a nearby cropland, *J. Geophys. Res. Biogeosci.*, 119, 722–740, doi:10.1002/2013JG002520.
- Chu, H., J. F. Gottgens, J. Chen, G. Sun, A. R. Desai, Z. Ouyang, C. Shao, and K. Czajkowski (2015), Climatic variability, hydrologic anomaly, and methane emission can turn productive freshwater marshes into net carbon sources, *Global Change Biol.*, 21(3), 1165–1181, doi:10.1111/gcb.12760.
- Cicerone, R. J., J. D. Shetter, and C. C. Delwiche (1983), Seasonal variation of methane flux from a California rice paddy, *J. Geophys. Res.*, 88(C15), 11,022–11,024, doi:10.1029/JC088C15p11022.
- Cicerone, R. J., C. C. Delwiche, S. C. Tyler, and P. R. Zimmerman (1992), Methane emissions from California rice paddies with varied treatments, *Global Biogeochem. Cycles*, 6(3), 233–248, doi:10.1029/92GB01412.
- Conrad, R. (2007), Microbial ecology of methanogens and methanotrophs, *Adv. Agron.*, 96, 1–63, doi:10.1016/S0065-2113(07)96005-8.
- Corton, T. M., J. B. Bajita, F. S. Grospe, R. R. Pamplona, C. A. Assis Jr., R. Wassmann, R. S. Lantin, and L. V. Buendia (2000), Methane emission from irrigated and intensively managed rice fields in Central Luzon (Philippines), *Nutr. Cycl. Agroecosyst.*, 58(1–3), 37–53, doi:10.1023/A:1009826131741.
- Dannenberg, S., and R. Conrad (1999), Effect of rice plants on methane production and rhizospheric metabolism in paddy soil, *Biogeochemistry*, 45(1), 53–71, doi:10.1007/BF00992873.
- Den van der Gon, H. A., and N. Van Breemen (1993), Diffusion-controlled transport of methane from soil to atmosphere as mediated by rice plants, *Biogeochemistry*, 21(3), 177–190, doi:10.1007/BF00001117.
- Dengel, S., D. Zona, T. Sachs, M. Aurela, M. Jammot, F. J. W. Parmentier, W. Oechel, and T. Vesala (2013), Testing the applicability of neural networks as a gap-filling method using CH₄ flux data from high latitude wetlands, *Biogeosciences*, 10(12), 8185–8200, doi:10.5194/bg-10-8185-2013.
- Denier van der Gon, H. A. C., P. M. van Bodegom, S. Houweling, P. H. Verburg, and N. van Breemen (2000), Combining upscaling and downscaling of methane emissions from rice fields: Methodologies and preliminary results, *Nutr. Cycl. Agroecosyst.*, 58(1), 285–301, doi:10.1023/a:1009863107151.
- Denier van der Gon, H. A. C., M. J. Kropff, N. van Breemen, R. Wassmann, R. S. Lantin, E. Aduna, T. M. Corton, and H. H. van Laar (2002), Optimizing grain yields reduces CH₄ emissions from rice paddy fields, *Proc. Natl. Acad. Sci. U.S.A.*, 99(19), 12,021–12,024, doi:10.1073/pnas.192276599.
- Detto, M., D. Baldocchi, and G. Katul (2010), Scaling properties of biologically active scalar concentration fluctuations in the atmospheric surface layer over a managed peatland, *Boundary Layer Meteorol.*, 136(3), 407–430, doi:10.1007/s10546-010-9514-z.
- Detto, M., J. Verfaillie, F. Anderson, L. Xu, and D. Baldocchi (2011), Comparing laser-based open- and closed-path gas analyzers to measure methane fluxes using the eddy covariance method, *Agric. For. Meteorol.*, 151(10), 1312–1324, doi:10.1016/j.agrformet.2011.05.014.
- Deverel, S. J., and D. A. Leighton (2010), Historic, recent, and future subsidence, Sacramento-San Joaquin Delta, California, USA, *San Francisco Estuary Watershed Sci.*, 8(2), 1–23.
- Deverel, S. J., and S. Rojstaczer (1996), Subsidence of agricultural lands in the Sacramento-San Joaquin Delta, California: Role of aqueous and gaseous carbon fluxes, *Water Resour. Res.*, 32(8), 2359–2367, doi:10.1029/96WR01338.
- Ding, A., C. Willis, R. Sass, and F. Fisher (1999), Methane emissions from rice fields: Effect of plant height among several rice cultivars, *Global Biogeochem. Cycles*, 13(4), 1045–1052, doi:10.1029/1999GB900084.
- Drexler, J. Z., R. L. Snyder, D. Spano, and U. K. T. Paw (2004), A review of models and micrometeorological methods used to estimate wetland evapotranspiration, *Hydrol. Processes*, 18(11), 2071–2101, doi:10.1002/hyp.1462.
- Drexler, J. Z., C. Fontaine, and S. Deverel (2009), The legacy of wetland drainage on the remaining peat in the Sacramento-San Joaquin Delta, California, USA, *Wetlands*, 29(1), 372–386, doi:10.1672/08-97.1.
- Eusufzai, M. K., T. Tokida, M. Okada, S.-i. Sugiyama, G. C. Liu, M. Nakajima, and R. Sameshima (2010), Methane emission from rice fields as affected by land use change, *Agric. Ecosyst. Environ.*, 139(4), 742–748, doi:10.1016/j.agee.2010.11.003.
- Foken, T., and B. Wichura (1996), Tools for quality assessment of surface-based flux measurements, *Agric. Ecosyst. Environ.*, 78(1–2), 83–105, doi:10.1016/0168-1923(95)02248-1.
- Griffin, D., and K. J. Anchukaitis (2014), How unusual is the 2012–2014 California drought? *Geophys. Res. Lett.*, 41, 9017–9023, doi:10.1002/2014GL02433.
- Grinsted, A., J. C. Moore, and S. Jevrejeva (2004), Application of the cross wavelet transform and wavelet coherence to geophysical time series, *Nonlinear Process. Geophys.*, 11(5/6), 561–566, doi:10.5194/npg-11-561-2004.

- Hatala, J. A., M. Detto, and D. D. Baldocchi (2012a), Gross ecosystem photosynthesis causes a diurnal pattern in methane emission from rice, *Geophys. Res. Lett.*, **39**, L06409, doi:10.1029/2012GL051303.
- Hatala, J. A., M. Detto, O. Sonnentag, S. J. Deverel, J. Verfaillie, and D. D. Baldocchi (2012b), Greenhouse gas (CO_2 , CH_4 , H_2O) fluxes from drained and flooded agricultural peatlands in the Sacramento-San Joaquin Delta, *Agric. Ecosyst. Environ.*, **150**, 1–18, doi:10.1016/j.agee.2012.01.009.
- Helfter, C., C. Campbell, K. J. Dinsmore, J. Drewer, M. Coyle, M. Anderson, U. Skiba, E. Nemitz, M. F. Billett, and M. A. Sutton (2015), Drivers of long-term variability in CO_2 net ecosystem exchange in a temperate peatland, *Biogeosciences*, **12**(6), 1799–1811, doi:10.5194/bg-12-1799-2015.
- Hendriks, D. M. D., J. van Huissteden, A. J. Dolman, and M. K. van der Molen (2007), The full greenhouse gas balance of an abandoned peat meadow, *Biogeosciences*, **4**(3), 411–424, doi:10.5194/bg-4-411-2007.
- Hendriks, D. M. D., J. van Huissteden, and A. J. Dolman (2010), Multi-technique assessment of spatial and temporal variability of methane fluxes in a peat meadow, *Agric. For. Meteorol.*, **150**(6), 757–774, doi:10.1016/j.agrformet.2009.06.017.
- Herbst, M., T. Friberg, K. Schelde, R. Jensen, R. Ringgaard, V. Vasquez, A. G. Thomsen, and H. Soegaard (2013), Climate and site management as driving factors for the atmospheric greenhouse gas exchange of a restored wetland, *Biogeosciences*, **10**, 39–52, doi:10.5194/bg-10-39-2013.
- Hollinger, S. E., C. J. Bernacchi, and T. P. Meyers (2005), Carbon budget of mature no-till ecosystem in North Central Region of the United States, *Agric. For. Meteorol.*, **130**(1–2), 59–69, doi:10.1016/j.agrformet.2005.01.005.
- Holzapfel-Pschorn, A., and W. Seiler (1986), Methane emission during a cultivation period from an Italian rice paddy, *J. Geophys. Res.*, **91**(D11), 11,803–11,814, doi:10.1029/JD091iD11p11803.
- Huang, Y. A. O., R. Sass, and F. Fisher (1997a), Methane emission from Texas rice paddy soils: 1. Quantitative multi-year dependence of CH_4 emission on soil, cultivar and grain yield, *Global Change Biol.*, **3**(6), 479–489, doi:10.1046/j.1365-2486.1997.00083.x.
- Huang, Y. A. O., R. Sass, and F. Fisher (1997b), Methane emission from Texas rice paddy soils: 2. Seasonal contribution of rice biomass production to CH_4 emission, *Global Change Biol.*, **3**(6), 491–500, doi:10.1046/j.1365-2486.1997.00106.x.
- Huang, Y. A. O., R. L. Sass, and J. F. M. Fisher (1998), A semi-empirical model of methane emission from flooded rice paddy soils, *Global Change Biol.*, **4**(3), 247–268, doi:10.1046/j.1365-2486.1998.00129.x.
- Humphreys, E. R. (2004), Net ecosystem production of three coastal Douglas-fir stands at different stages of development after harvesting, Univ. of British Columbia.
- IPCC (1997), *Revised 1996 IPCC Guidelines for National Greenhouse Gas Inventories*, Cambridge Univ. Press, Cambridge.
- Katul, G. G., J. Schieldge, C.-I. Hsieh, and B. Vidakovic (1998), Skin temperature perturbations induced by surface layer turbulence above a grass surface, *Water Resour. Res.*, **34**(5), 1265–1274, doi:10.1029/98WR00293.
- Khalil, M. A. K., and C. L. Butenhoff (2008), Spatial variability of methane emissions from rice fields and implications for experimental design, *J. Geophys. Res.*, **113**, G00A09, doi:10.1029/2007JG000517.
- Khalil, M. A. K., R. A. Rasmussen, M. J. Shearer, R. W. Dalluge, L. Ren, and C.-L. Duan (1998), Factors affecting methane emissions from rice fields, *J. Geophys. Res.*, **103**(D19), 25,219–25,231, doi:10.1029/98JD01115.
- Knox, S. H., C. Sturtevant, J. H. Matthes, L. Koteen, J. Verfaillie, and D. Baldocchi (2014), Agricultural peatland restoration: Effects of land-use change on greenhouse gas (CO_2 and CH_4) fluxes in the Sacramento-San Joaquin Delta, *Global Change Biol.*, **21**, 750–765, doi:10.1111/gcb.12745.
- Koebisch, F., G. Jurasinski, M. Koch, J. Hofmann, and S. Glatzel (2015), Controls for multi-scale temporal variation in ecosystem methane exchange during the growing season of a permanently inundated fen, *Agric. For. Meteorol.*, **204**, 94–105, doi:10.1016/j.agrformet.2015.02.002.
- Kramer, R. A., and L. Shabman (1993), The effects of agricultural and tax policy reform on the economic return to wetland drainage in the Mississippi Delta region, *Land Econ.*, **69**(3), 249–262, doi:10.2307/3146591.
- Kroon, P. S., A. P. Schrier-Uijl, A. Hensen, E. M. Veenendaal, and H. J. J. Jonker (2010), Annual balances of CH_4 and N_2O from a managed fen meadow using eddy covariance flux measurements, *Eur. J. Soil Sci.*, **61**(5), 773–784, doi:10.1111/j.1365-2389.2010.01273.x.
- Le Mer, J., and P. Roger (2001), Production, oxidation, emission and consumption of methane by soils: A review, *Eur. J. Soil Biol.*, **37**(1), 25–50, doi:10.1016/S1164-5563(01)01067-6.
- Leuning, R., E. van Gorsel, W. J. Massman, and P. R. Isaac (2012), Reflections on the surface energy imbalance problem, *Agric. For. Meteorol.*, **156**, 65–74, doi:10.1016/j.agrformet.2011.12.002.
- Lindau, C. W., P. K. Bollich, R. D. Delaune, W. H. Patrick Jr., and V. J. Law (1991), Effect of urea fertilizer and environmental factors on CH_4 emissions from a Louisiana, USA rice field, *Plant Soil*, **136**(2), 195–203, doi:10.1007/BF02150050.
- Linguist, B. A., M. M. Anders, M. A. A. Adviento-Borbe, R. L. Chaney, L. L. Nalley, E. F. F. da Rosa, and C. van Kessel (2015), Reducing greenhouse gas emissions, water use, and grain arsenic levels in rice systems, *Global Change Biol.*, **21**(1), 407–417, doi:10.1111/gcb.12701.
- Malone, S. L., C. L. Staudhammer, H. W. Loescher, P. Olivas, S. F. Oberbauer, M. G. Ryan, J. Schedlbauer, and G. Starr (2014), Seasonal patterns in energy partitioning of two freshwater marsh ecosystems in the Florida Everglades, *J. Geophys. Res. Biogeosci.*, **119**, 1487–1505, doi:10.1002/2014JG002700.
- Massman, W. J., and X. Lee (2002), Eddy covariance flux corrections and uncertainties in long-term studies of carbon and energy exchanges, *Agric. For. Meteorol.*, **113**(1–4), 121–144, doi:10.1016/S0168-1923(02)00105-3.
- Matthes, J. H., C. Sturtevant, J. Verfaillie, S. Knox, and D. Baldocchi (2014), Parsing the variability in CH_4 flux at a spatially heterogeneous wetland: Integrating multiple eddy covariance towers with high-resolution flux footprint analysis, *J. Geophys. Res. Biogeosci.*, **119**, 1322–1339, doi:10.1002/2014JG002642.
- McDermitt, D., et al. (2011), A new low-power, open-path instrument for measuring methane flux by eddy covariance, *Appl. Phys. B*, **102**(2), 391–405, doi:10.1007/s00340-010-4307-0.
- McMillan, A. M. S., M. L. Goulden, and S. C. Tyler (2007), Stoichiometry of CH_4 and CO_2 flux in a California rice paddy, *J. Geophys. Res.*, **112**, G01008, doi:10.1029/2006JG000198.
- Meijide, A., G. Manca, I. Goded, V. Magliulo, P. di Tommasi, G. Seufert, and A. Cescatti (2011), Seasonal trends and environmental controls of methane emissions in a rice paddy field in Northern Italy, *Biogeosciences*, **8**(12), 3809–3821, doi:10.5194/bg-8-3809-2011.
- Minoda, T., and M. Kimura (1994), Contribution of photosynthesized carbon to the methane emitted from paddy fields, *Geophys. Res. Lett.*, **21**(18), 2007–2010, doi:10.1029/94GL01595.
- Minoda, T., M. Kimura, and E. Wada (1996), Photosynthates as dominant source of CH_4 and CO_2 in soil water and CH_4 emitted to the atmosphere from paddy fields, *J. Geophys. Res.*, **101**(D15), 21,091–21,097, doi:10.1029/96JD01710.
- Miyata, A., T. Iwata, H. Nagai, T. Yamada, H. Yoshikoshi, M. Mano, K. Ono, G. Han, Y. Harazono, and E. Ohtaki (2005), Seasonal variation of carbon dioxide and methane fluxes at single cropping paddy fields in central and western Japan, *Phyton*, **45**(4), 89–97.
- Moffat, A. M., et al. (2007), Comprehensive comparison of gap-filling techniques for eddy covariance net carbon fluxes, *Agric. For. Meteorol.*, **147**(3–4), 209–232, doi:10.1016/j.agrformet.2007.08.011.

- Monteith, J. (1969), Light interception and radiative exchange in crop stands, paper presented at Physiological Aspects of Crop Yield: Proceedings of a Symposium Sponsored by the University of Nebraska, the American Society of Agronomy, and the Crop Science Society of America, University of Nebraska, Lincoln.
- Moore, C. J. (1986), Frequency response corrections for eddy correlation systems, *Boundary Layer Meteorol.*, 37(1–2), 17–35, doi:10.1007/BF00122754.
- Morin, T. H., G. Bohrer, R. P. d. M. Frasson, L. Naor-Azreli, S. Mesj, K. C. Stefanik, and K. V. R. Schäfer (2014), Environmental drivers of methane fluxes from an urban temperate wetland park, *J. Geophys. Res. Biogeosci.*, 119, 2188–2208, doi:10.1002/2014JG002750.
- Mount, J., and R. Twiss (2005), Subsidence, sea level rise, and seismicity in the Sacramento-San Joaquin Delta, *San Francisco Estuary Watershed Sci.*, 3(1), 1–18.
- Myhre, G., et al. (2013), Anthropogenic and natural radiative forcing, in *Climate Change 2013: The Physical Science Basis. Contribution of Working Group I to the Fifth Assessment Report of the Intergovernmental Panel on Climate Change*, edited by T. F. Stocker et al., pp. 659–740, Cambridge Univ. Press, U. K., and New York, doi:10.1017/CBO9781107415324.018.
- Naser, H. M., O. Nagata, S. Tamura, and R. Hatano (2007), Methane emissions from five paddy fields with different amounts of rice straw application in central Hokkaido, Japan, *Soil Sci. Plant Nutr.*, 53(1), 95–101, doi:10.1111/j.1747-0765.2007.00105.x.
- Neue, H. U., and R. Sass (1994), Trace gas emissions from rice fields, in *Global Atmospheric-Biospheric Chemistry*, edited by R. Prinn, pp. 119–147, Springer, New York, doi:10.1007/978-1-4615-2524-0_8.
- Neue, H. U., R. Wassmann, H. K. Kludze, W. Bujun, and R. S. Lantin (1997), Factors and processes controlling methane emissions from rice fields, *Nutr. Cycl. Agroecosyst.*, 49(1–3), 111–117, doi:10.1023/A:1009714526204.
- Nishimura, S., T. Sawamoto, H. Akiyama, S. Sudo, and K. Yagi (2004), Methane and nitrous oxide emissions from a paddy field with Japanese conventional water management and fertilizer application, *Global Biogeochem. Cycles*, 18 GB2017, doi:10.1029/2003GB002207.
- Nouchi, I., and S. Mariko (1993), Mechanism of methane transport by rice plants, in *Biogeochemistry of Global Change*, edited by R. Oremland, pp. 336–352, Springer, New York, doi:10.1007/978-1-4615-2812-8_18.
- Novick, K. A., A. C. Oishi, E. J. Ward, M. B. S. Siqueira, J.-Y. Juang, and P. C. Stoy (2015), On the difference in the net ecosystem exchange of CO₂ between deciduous and evergreen forests in the southeastern United States, *Global Change Biol.*, 21(2), 827–842, doi:10.1111/gcb.12723.
- Papale, D., et al. (2006), Towards a standardized processing of Net Ecosystem Exchange measured with eddy covariance technique: Algorithms and uncertainty estimation, *Biogeosciences*, 3(4), 571–583, doi:10.5194/bg-3-571-2006.
- Petrescu, A. M. R., et al. (2015), The uncertain climate footprint of wetlands under human pressure, *Proc. Natl. Acad. Sci. U.S.A.*, 112(15), 4594–4599, doi:10.1073/pnas.1416267112.
- Reeburgh, W. (2003), Global methane biogeochemistry, *Treatise Geochem.*, 4, 65–89, doi:10.1016/B0-08-043751-6/04036-6.
- Reichstein, M., et al. (2005), On the separation of net ecosystem exchange into assimilation and ecosystem respiration: Review and improved algorithm, *Global Change Biol.*, 11(9), 1424–1439, doi:10.1111/j.1365-2486.2005.001002.x.
- Richardson, A., J. Jenkins, B. Braswell, D. Hollinger, S. Ollinger, and M.-L. Smith (2007), Use of digital webcam images to track spring green-up in a deciduous broadleaf forest, *Oecologia*, 152(2), 323–334, doi:10.1007/s00442-006-0657-z.
- Rinne, J., T. Riutta, M. Pihlatie, M. Aurela, S. Haapanala, J.-P. Tuovinen, E.-S. Tuittila, and T. Vesala (2007), Annual cycle of methane emission from a boreal fen measured by the eddy covariance technique, *Tellus B*, 59(3), 449–457, doi:10.1111/j.1600-0889.2007.00261.x.
- Rocha, A. V., and M. L. Goulden (2008), Large interannual CO₂ and energy exchange variability in a freshwater marsh under consistent environmental conditions, *J. Geophys. Res.*, 113, G04019, doi:10.1029/2008JG000712.
- Roulet, N. T., P. M. Lafleur, P. J. H. Richard, T. R. Moore, E. R. Humphreys, and J. Bubier (2007), Contemporary carbon balance and late Holocene carbon accumulation in a northern peatland, *Global Change Biol.*, 13(2), 397–411, doi:10.1111/j.1365-2486.2006.01292.x.
- Ruimy, A., L. Kergoat, A. Bondeau, and Intercomparison, ThE. P. Of. ThE. P. NpP. M (1999), Comparing global models of terrestrial net primary productivity (NPP): Analysis of differences in light absorption and light-use efficiency, *Global Change Biol.*, 5(5), 56–64, doi:10.1046/j.1365-2486.1999.00007.x.
- Saito, M., A. Miyata, H. Nagai, and T. Yamada (2005), Seasonal variation of carbon dioxide exchange in rice paddy field in Japan, *Agric. For. Meteorol.*, 135(1–4), 93–109, doi:10.1016/j.agrformet.2005.10.007.
- Sass, R. L., F. M. Fisher, P. A. Harcombe, and F. T. Turner (1990), Methane production and emission in a Texas rice field, *Global Biogeochem. Cycles*, 4(1), 47–68, doi:10.1029/GB004i001p00047.
- Sass, R. L., F. M. Fisher, P. A. Harcombe, and F. T. Turner (1991a), Mitigation of methane emissions from rice fields: Possible adverse effects of incorporated rice straw, *Global Biogeochem. Cycles*, 5(3), 275–287, doi:10.1029/91GB01304.
- Sass, R. L., F. M. Fisher, F. T. Turner, and M. F. Jund (1991b), Methane emission from rice fields as influenced by solar radiation, temperature, and straw incorporation, *Global Biogeochem. Cycles*, 5(4), 335–350, doi:10.1029/91GB02586.
- Sass, R. L., F. M. Fisher, S. T. Lewis, M. F. Jund, and F. T. Turner (1994), Methane emissions from rice fields: Effect of soil properties, *Global Biogeochem. Cycles*, 8(2), 135–140, doi:10.1029/94GB00588.
- Sass, R. L., F. M. Fisher, Jr., and Y. Huang (2000), A process-based model for methane emissions from irrigated rice fields: Experimental basis and assumptions, in *Methane Emissions From Major Rice Ecosystems in Asia*, edited by R. Wassmann, R. Lantin, and H.-U. Neue, pp. 249–258, Springer, Netherlands, doi:10.1007/978-94-010-0898-3_19.
- Satpathy, S. N., A. K. Rath, B. Ramakrishnan, V. R. Rao, T. K. Adhya, and N. Sethunathan (1997), Diurnal variation in methane efflux at different growth stages of tropical rice, *Plant Soil*, 195(2), 267–271, doi:10.1023/A:1004202515767.
- Schütz, H., A. Holzapfel-Pschorn, R. Conrad, H. Rennenberg, and W. Seiler (1989a), A 3-year continuous record on the influence of daytime, season, and fertilizer treatment on methane emission rates from an Italian rice paddy, *J. Geophys. Res.*, 94(D13), 16,405–16,416, doi:10.1029/JD094iD13p16405.
- Schütz, H., W. Seiler, and R. Conrad (1989b), Processes involved in formation and emission of methane in rice paddies, *Biogeochemistry*, 7(1), 33–53, doi:10.1007/BF00000896.
- Schütz, H., W. Seiler, and R. Conrad (1990), Influence of soil temperature on methane emission from rice paddy fields, *Biogeochemistry*, 11(2), 77–95, doi:10.1007/BF00002060.
- Seiler, W., A. Holzapfel-Pschorn, R. Conrad, and D. Scharffe (1983), Methane emission from rice paddies, *J. Atmos. Chem.*, 1(3), 241–268, doi:10.1007/BF00058731.
- Stallard, R. F. (1998), Terrestrial sedimentation and the carbon cycle: Coupling weathering and erosion to carbon burial, *Global Biogeochem. Cycles*, 12(2), 231–257, doi:10.1029/98GB00741.
- Stoy, P. C., G. G. Katul, M. B. S. Siqueira, J.-Y. Juang, H. R. McCarthy, H.-S. Kim, A. C. Oishi, and R. Oren (2005), Variability in net ecosystem exchange from hourly to inter-annual time scales at adjacent pine and hardwood forests: A wavelet analysis, *Tree Physiol.*, 25(7), 887–902, doi:10.1093/treephys/25.7.887.

- Stoy, P. C., et al. (2013), A data-driven analysis of energy balance closure across FLUXNET research sites: The role of landscape scale heterogeneity, *Agric. For. Meteorol.*, 171–172, 137–152, doi:10.1016/j.agrformet.2012.11.004.
- Strachan, I., K. Nugent, S. Crombie, and M. C. Bonneville (2015), Carbon dioxide and methane exchange at a cool-temperate freshwater marsh, *Environ. Res. Lett.*, 10(6), doi:10.1088/1748-9326/10/6/065006.
- Ström, L., A. Ekberg, M. Mastepanov, and T. Røjle Christensen (2003), The effect of vascular plants on carbon turnover and methane emissions from a tundra wetland, *Global Change Biol.*, 9(8), 1185–1192, doi:10.1046/j.1365-2486.2003.00655.x.
- Sturtevant, C., and W. C. Oechel (2013), Spatial variation in landscape-level CO₂ and CH₄ fluxes from arctic coastal tundra: Influence from vegetation, wetness, and the thaw lake cycle, *Global Change Biol.*, 19(9), 2853–2866, doi:10.1111/gcb.12247.
- Sturtevant, C., W. C. Oechel, D. Zona, Y. Kim, and C. E. Emerson (2012), Soil moisture control over autumn season methane flux, Arctic Coastal Plain of Alaska, *Biogeosciences*, 9(4), 1423–1440, doi:10.5194/bg-9-1423-2012.
- Sturtevant, C., B. L. Ruddell, S. H. Knox, J. Verfaillie, J. H. Matthes, P. Y. Oikawa, and D. Baldocchi (2015), Identifying scale-emergent, non-linear, asynchronous processes of wetland methane exchange, *J. Geophys. Res. Biogeosci.*, 121, 188–204, doi:10.1002/2015JG003054.
- Taylor, J. A., G. P. Brasseur, P. R. Zimmerman, and R. J. Cicerone (1991), A study of the sources and sinks of methane and methyl chloroform using a global three-dimensional Lagrangian tropospheric tracer transport model, *J. Geophys. Res.*, 96(D2), 3013–3044, doi:10.1029/90JD02016.
- Torrence, C., and G. P. Compo (1998), A practical guide to wavelet analysis, *Bull. Am. Meteorol. Soc.*, 79(1), 61–78, doi:10.1175/1520-0477(1998)079<0061:APGTWA>2.0.CO;2.
- Treat, C. C., J. L. Bubier, R. K. Varner, and P. M. Crill (2007), Timescale dependence of environmental and plant-mediated controls on CH₄ flux in a temperate fen, *J. Geophys. Res.*, 112 G01014, doi:10.1029/2006JG000210.
- Wang, B., H. U. Neue, and H. P. Samonte (1997), Role of rice in mediating methane emission, *Plant Soil*, 189(1), 107–115, doi:10.1023/A:1004219024281.
- Wassmann, R., and M. S. Aulakh (2000), The role of rice plants in regulating mechanisms of methane emissions, *Biol. Fertil. Soils*, 31(1), 20–29, doi:10.1007/s003740050619.
- Wassmann, R., R. S. Lantin, H. U. Neue, L. V. Buendia, T. M. Corton, and Y. Lu (2000a), Characterization of methane emissions from rice fields in Asia: III. Mitigation options and future research needs, *Nutr. Cycl. Agroecosyst.*, 58(1–3), 23–36, doi:10.1023/A:1009874014903.
- Wassmann, R., H. U. Neue, R. S. Lantin, L. V. Buendia, and H. Rennenberg (2000b), Characterization of methane emissions from rice fields in Asia: I. Comparison among field sites in five countries, *Nutr. Cycl. Agroecosyst.*, 58(1–3), 1–12, doi:10.1023/A:1009848813994.
- Whalen, S. (2005), Biogeochemistry of methane exchange between natural wetlands and the atmosphere, *Environ. Eng. Sci.*, 22(1), 73–94, doi:10.1089/ees.2005.22.73.
- Whiting, G. J., and J. P. Chanton (1993), Primary production control of methane emission from wetlands, *Nature*, 364(6440), 794–795, doi:10.1038/364794a0.
- Wilson, K., et al. (2002), Energy balance closure at FLUXNET sites, *Agric. For. Meteorol.*, 113(1–4), 223–243, doi:10.1016/S0168-1923(02)00109-0.
- Yagi, K., and K. Minami (1990), Effect of organic matter application on methane emission from some Japanese paddy fields, *Soil Sci. Plant Nutr.*, 36(4), 599–610, doi:10.1080/00380768.1990.10416797.
- Yagi, K., and K. Minami (1993), Spatial and temporal variations of methane flux from a rice paddy field, in *Biogeochemistry of Global Change*, edited by R. Oremland, pp. 353–368, Springer, New York, doi:10.1007/978-1-4615-2812-8_19.
- Yagi, K., H. Tsuruta, K. I. Kanda, and K. Minami (1996), Effect of water management on methane emission from a Japanese rice paddy field: Automated methane monitoring, *Global Biogeochem. Cycles*, 10(2), 255–267, doi:10.1029/96GB00517.
- Yan, X., K. Yagi, H. Akiyama, and H. Akimoto (2005), Statistical analysis of the major variables controlling methane emission from rice fields, *Global Change Biol.*, 11(7), 1131–1141, doi:10.1111/j.1365-2486.2005.00976.x.
- Ye, R., T. A. Doane, J. Morris, and W. R. Horwath (2015), The effect of rice straw on the priming of soil organic matter and methane production in peat soils, *Soil Biol. Biochem.*, 81, 98–107, doi:10.1016/j.soilbio.2014.11.007.
- Ye, R., M. B. Espe, B. Linquist, S. J. Parikh, T. A. Doane, and W. R. Horwath (2016), A soil carbon proxy to predict CH₄ and N₂O emissions from rewetted agricultural peatlands, *Agric. Ecosyst. Environ.*, 220, 64–75, doi:10.1016/j.agee.2016.01.008.
- Yvon-Durocher, G., A. P. Allen, D. Bastviken, R. Conrad, C. Gudas, A. St-Pierre, N. Thanh-Duc, and P. A. del Giorgio (2014), Methane fluxes show consistent temperature dependence across microbial to ecosystem scales, *Nature*, 507(7493), 488–491, doi:10.1038/nature13164.

Erratum

In the originally published version of this article, the Supporting Information was from an unrelated paper. The correct Supporting Information has since been uploaded and this version may be considered the authoritative version of record.